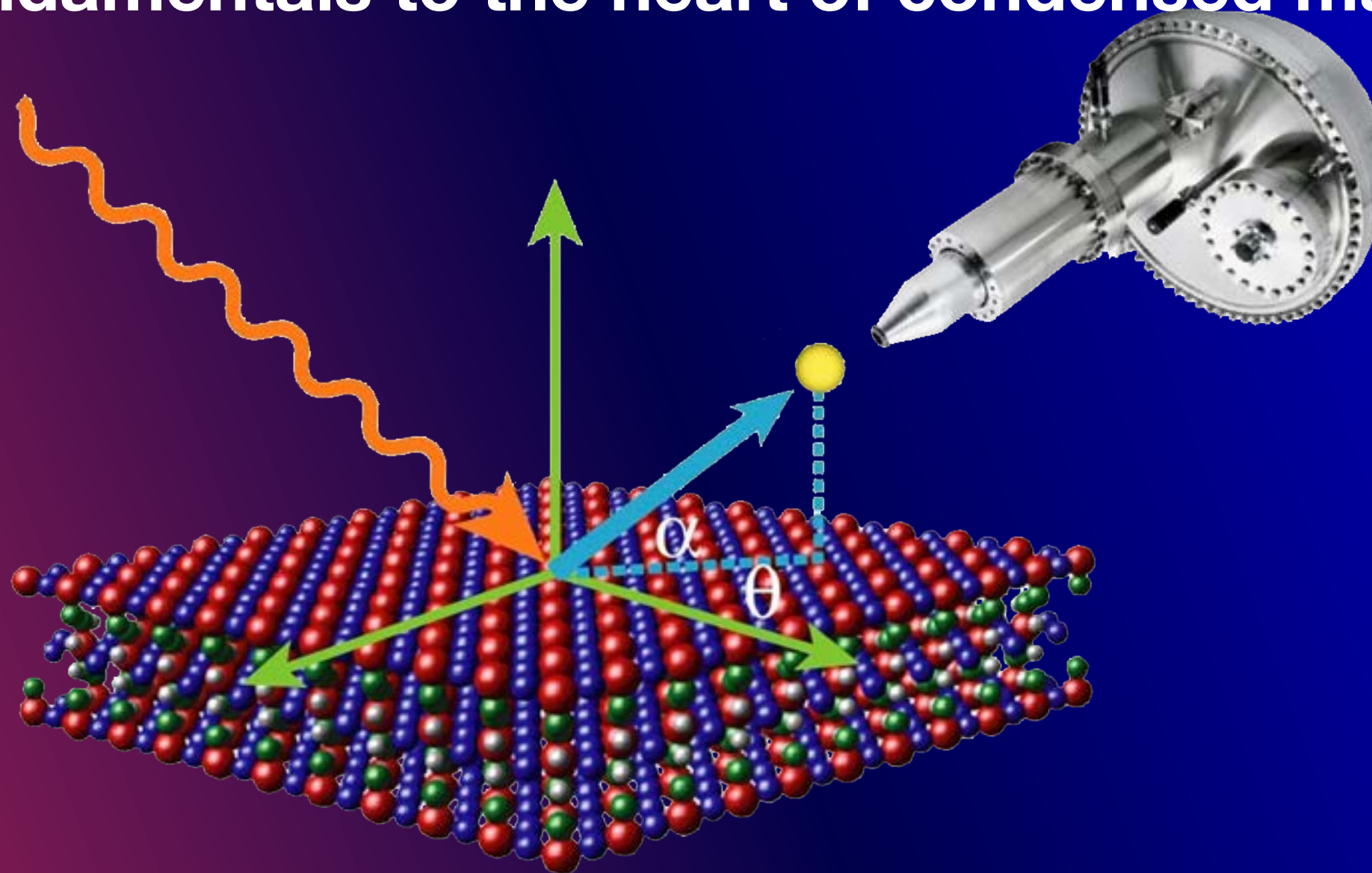


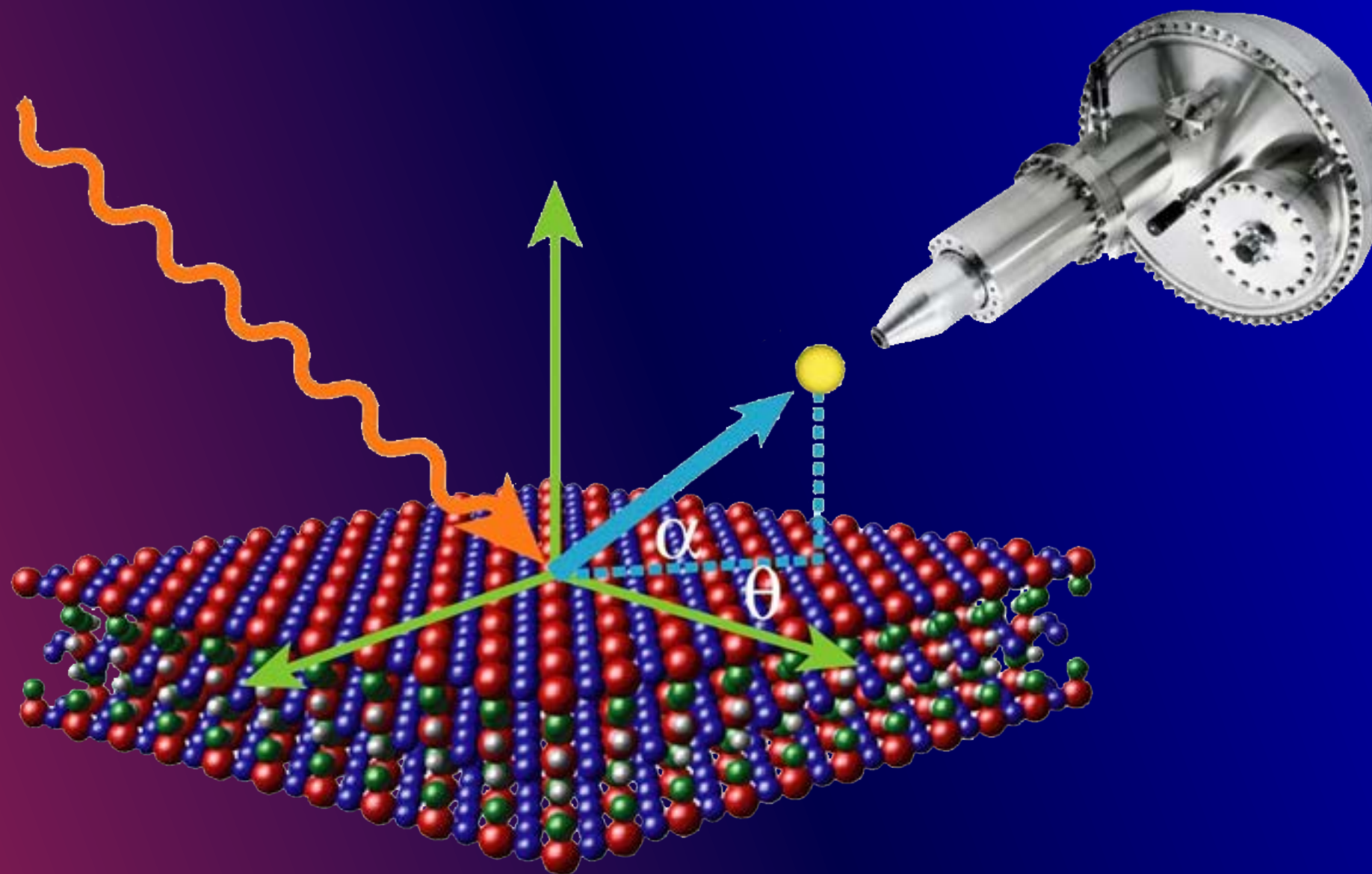
Angle Resolved Photoemission Spectroscopy

From fundamentals to the heart of condensed matter



6-7 FEBRERO, 2023

Lecture #3 : ARPES Studies of Quantum Materials



6-7 FEBRERO, 2023

Consider an ARPES experiment being conducting with an electron analyzer resolution of $\Delta E = 10$ meV and an photon bandwidth of $\Delta E = 2$ meV

What is the closest value of the **TOTAL** “effective” energy broadening in the experiment?

- A. 10 meV
- B. 11 meV
- C. 12 meV
- D. 13 meV

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In your ARPES experiment of a given material, you see a sharp “kink” in the quasiparticle band dispersion, which is a clear signature of some electron-boson interaction.

What information **cannot** be directly obtained from the analysis of your experimental data?

- A. The energy of the boson
- B. The strength of the electron-boson interaction
- C. What kind of boson it is (i.e. phonon, magnon, etc...)
- D. None of A, B, and C can be obtained from the data
- E. All of A, B, and C can be obtained from the data

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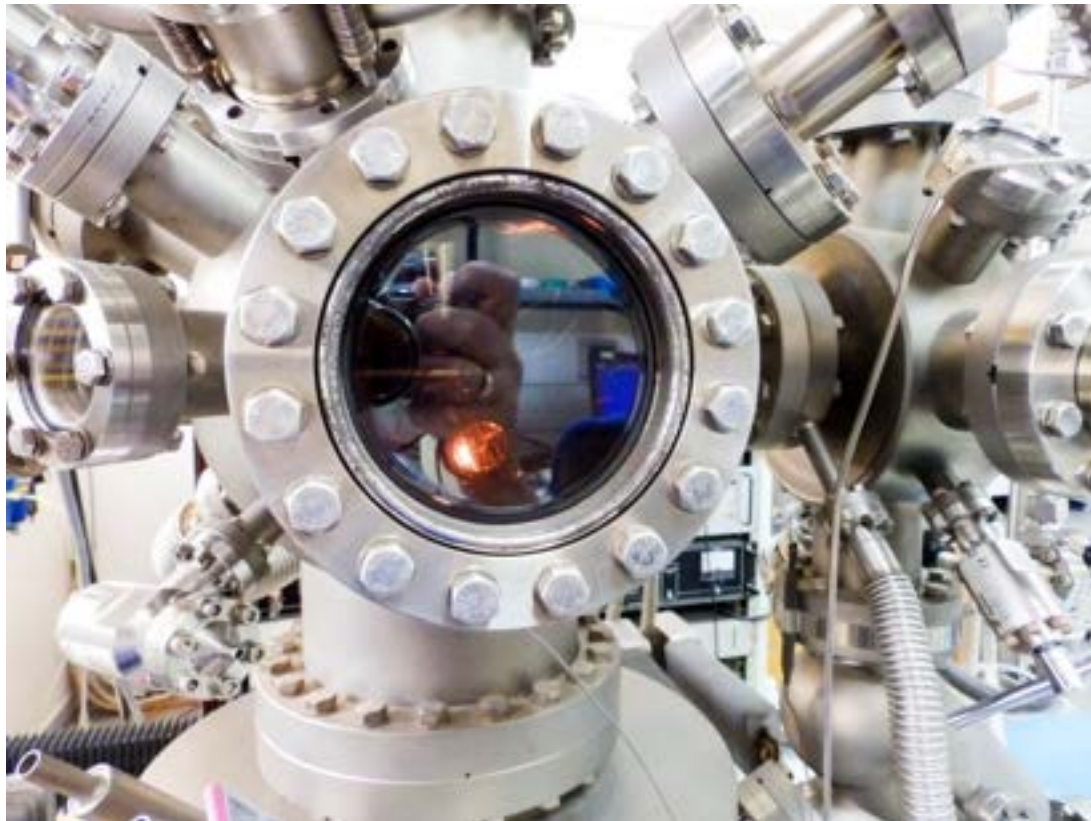
Which of the following material properties would be directly observable using ARPES?

- | | |
|--|---------------------|
| 1. The exciton binding energy in a semiconductor | A. 2 & 4 |
| 2. The strength of electron-phonon interactions in a metal | B. 1, 3, 4, & 5 |
| 3. The magnon (spin-wave) dispersion in an antiferromagnet | C. 2, 3, 4, & 5 |
| 4. The interacting quasiparticle band structure | D. 3, 4, & 5 |
| 5. The electronic band structure as calculated by DFT | E. All of the above |

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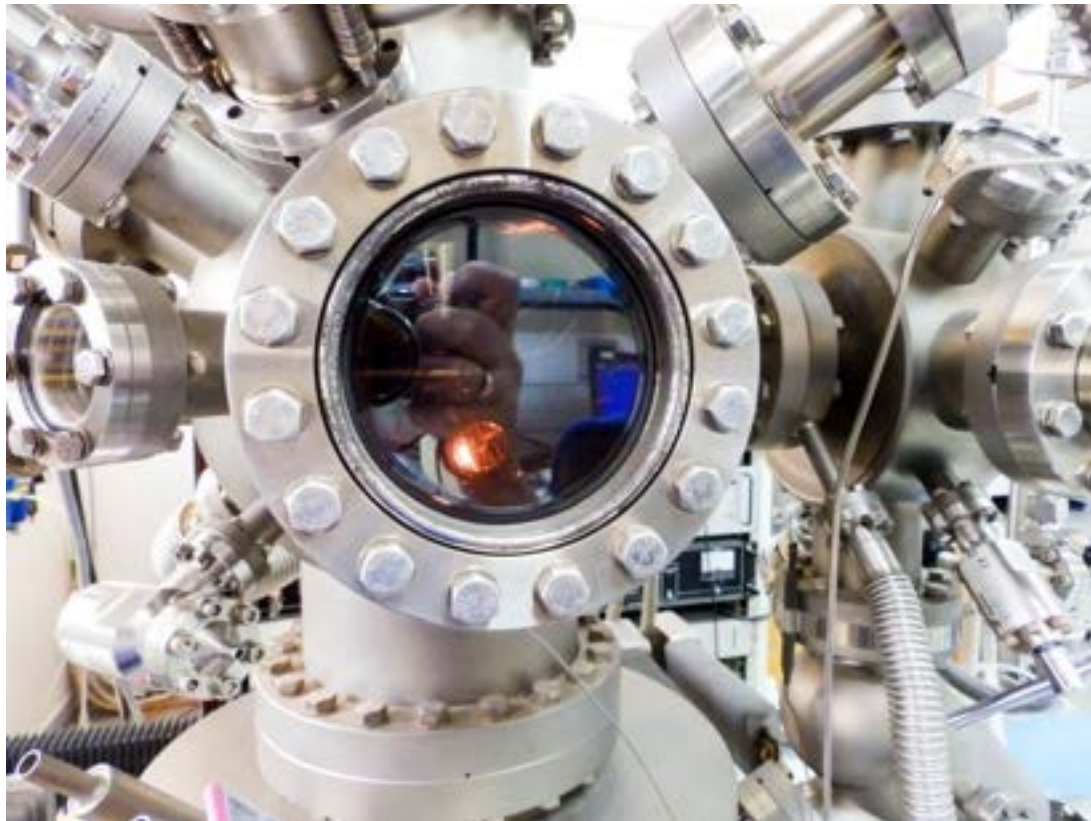
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ARPES measurements need to take place in ultrahigh vacuum (10^{-10} torr or better). Which of the following is the **most important** factor which determines the level of vacuum needed to perform experiments?



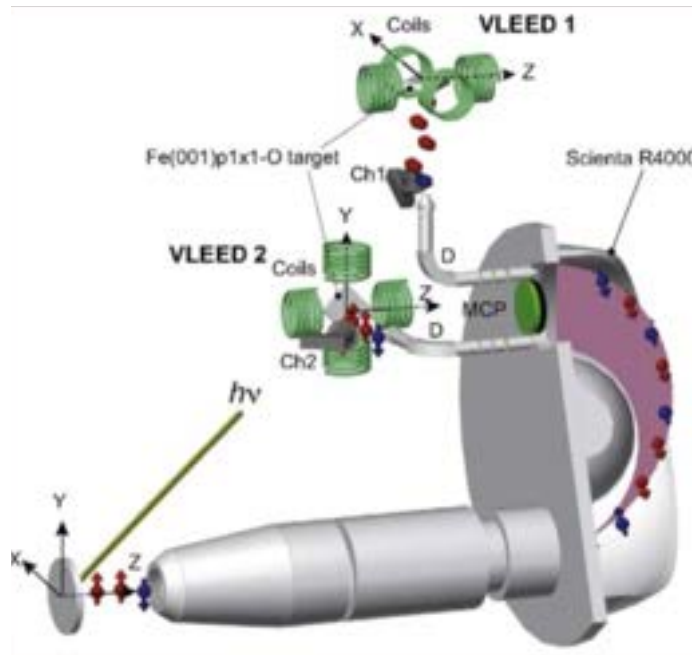
- A. The scattering / absorption of photoelectrons traveling inside the chamber
- B. The operation of the electron analyzer
- C. The absorption of vacuum ultraviolet (VUV) photons used for photoemitting the electrons
- D. The scattering of electrons from adsorbed molecules at the sample's surface
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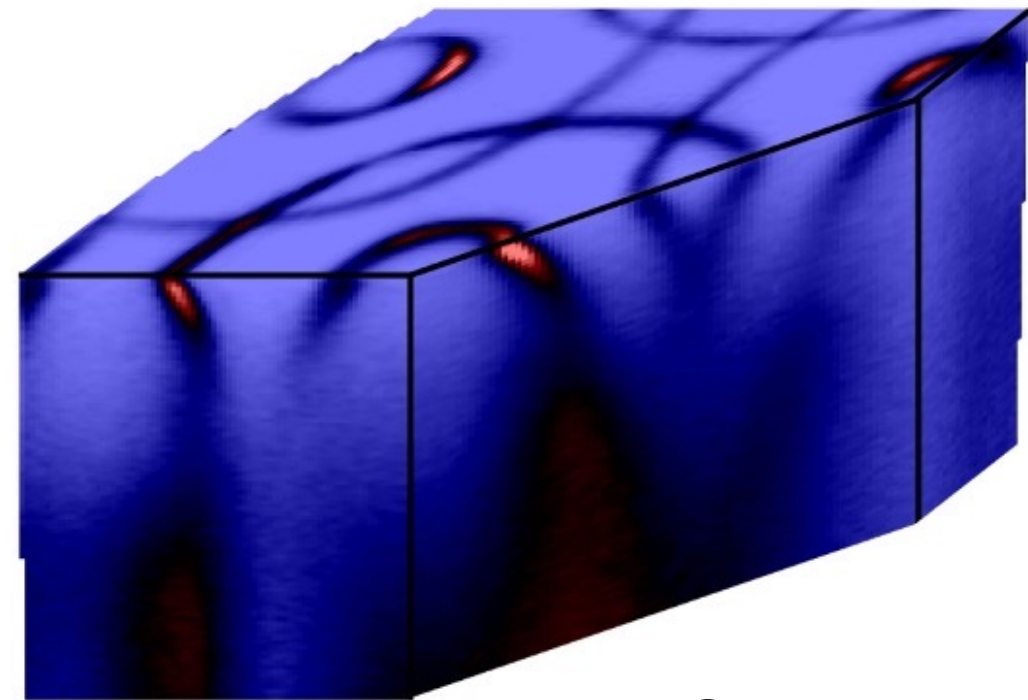


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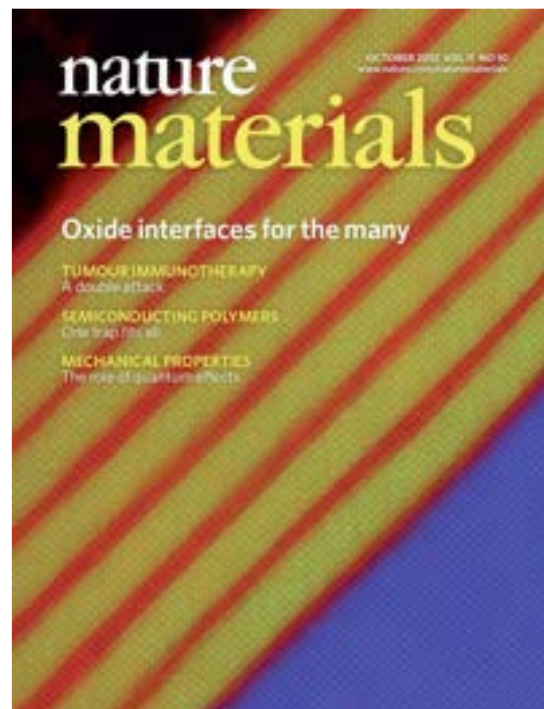
New Instrumentation



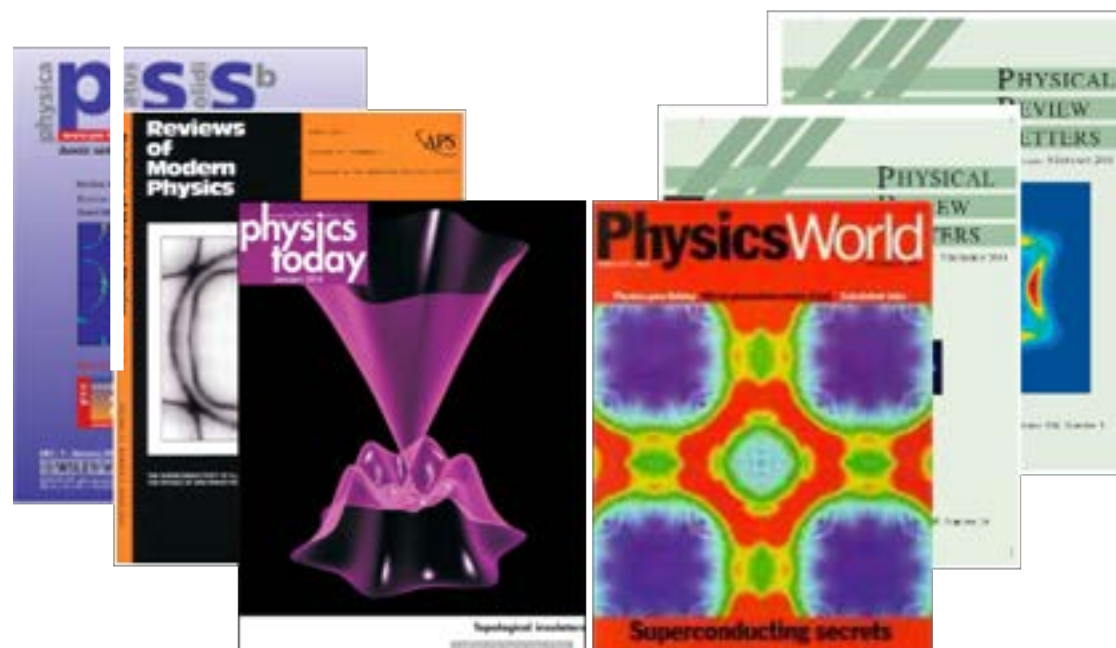
Data Processing



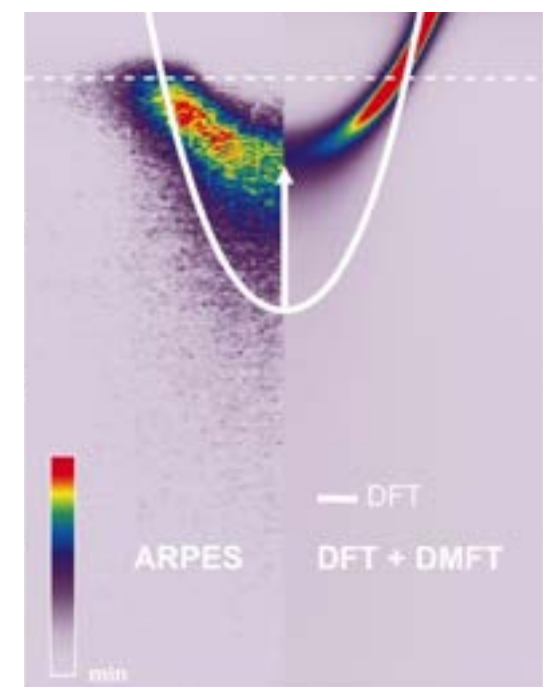
New Materials

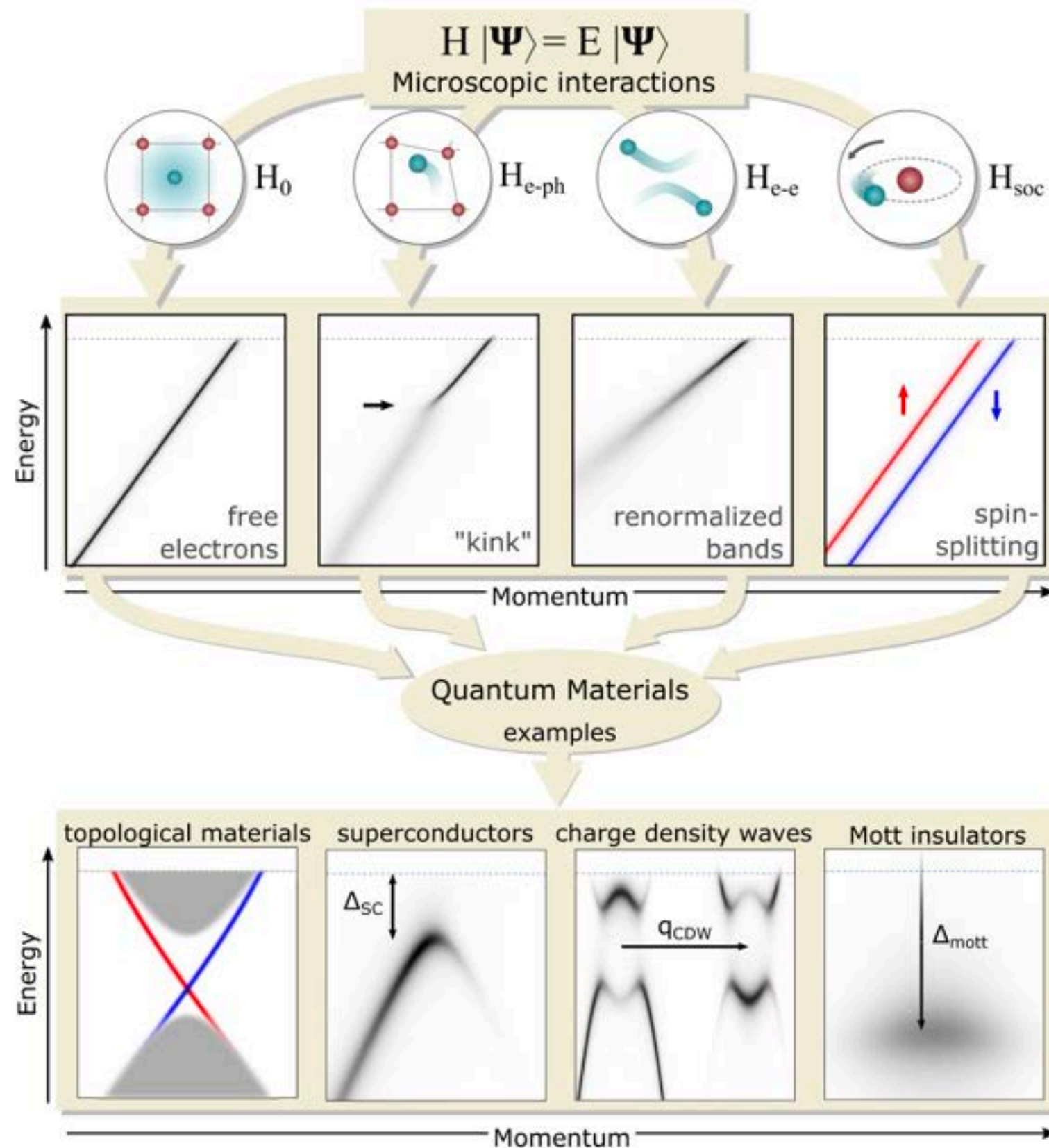


Scientific Questions



State-of-the-art Theory





Example #1 : High-Temperature Cuprate Superconductors

- Evolution from the parent Mott insulating state
- *d*-wave superconducting gap
- discovery of the pseudogap

Example #2 : Strain Engineering of Superconductivity

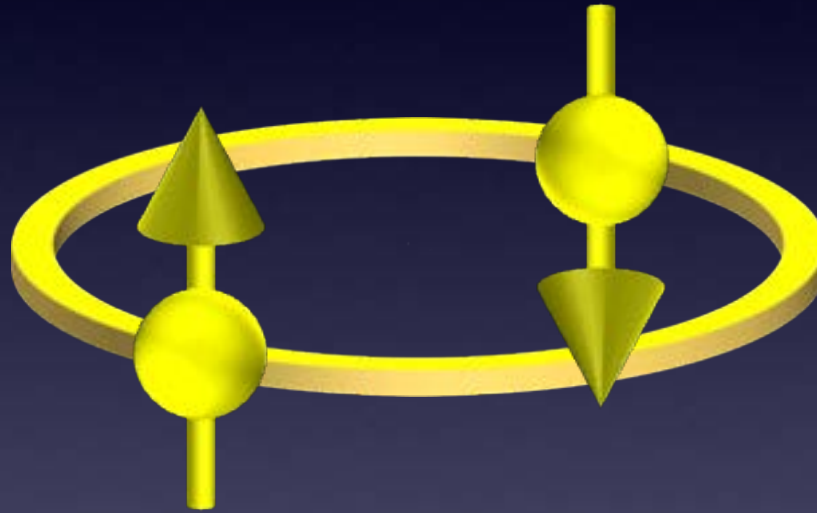
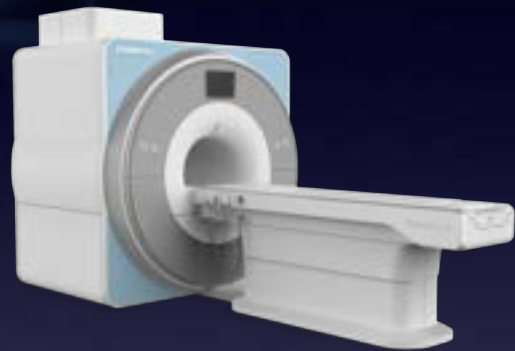
- “Fermi Surface Engineering” in strained Sr_2RuO_4
- The first strain-stabilized superconductor : RuO_2

Example #3 : Monolayer High-Tc Interfacial Superconductivity

- Pairing enhancement in $\text{FeSe} / \text{SrTiO}_3$
- Interfacial Electron-Phonon Coupling in $\text{FeSe} / \text{SrTiO}_3$

superconductivity : macroscopic quantum phenomena

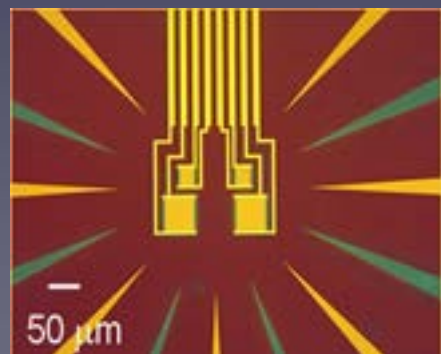
magnetic fields



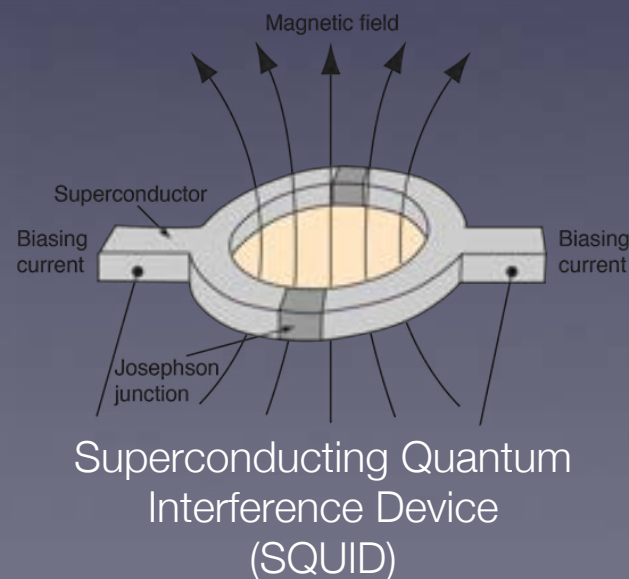
energy applications



quantum sensors

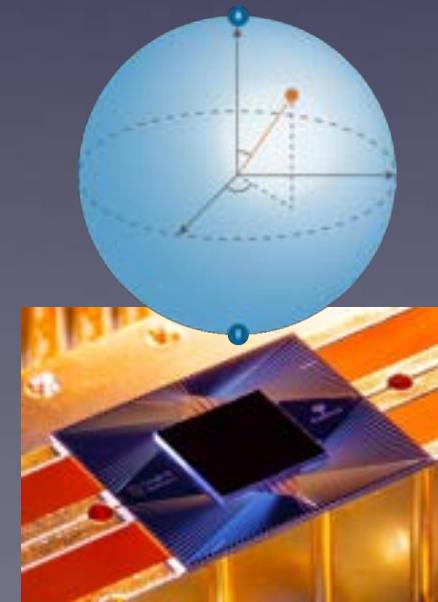


transition edge sensor

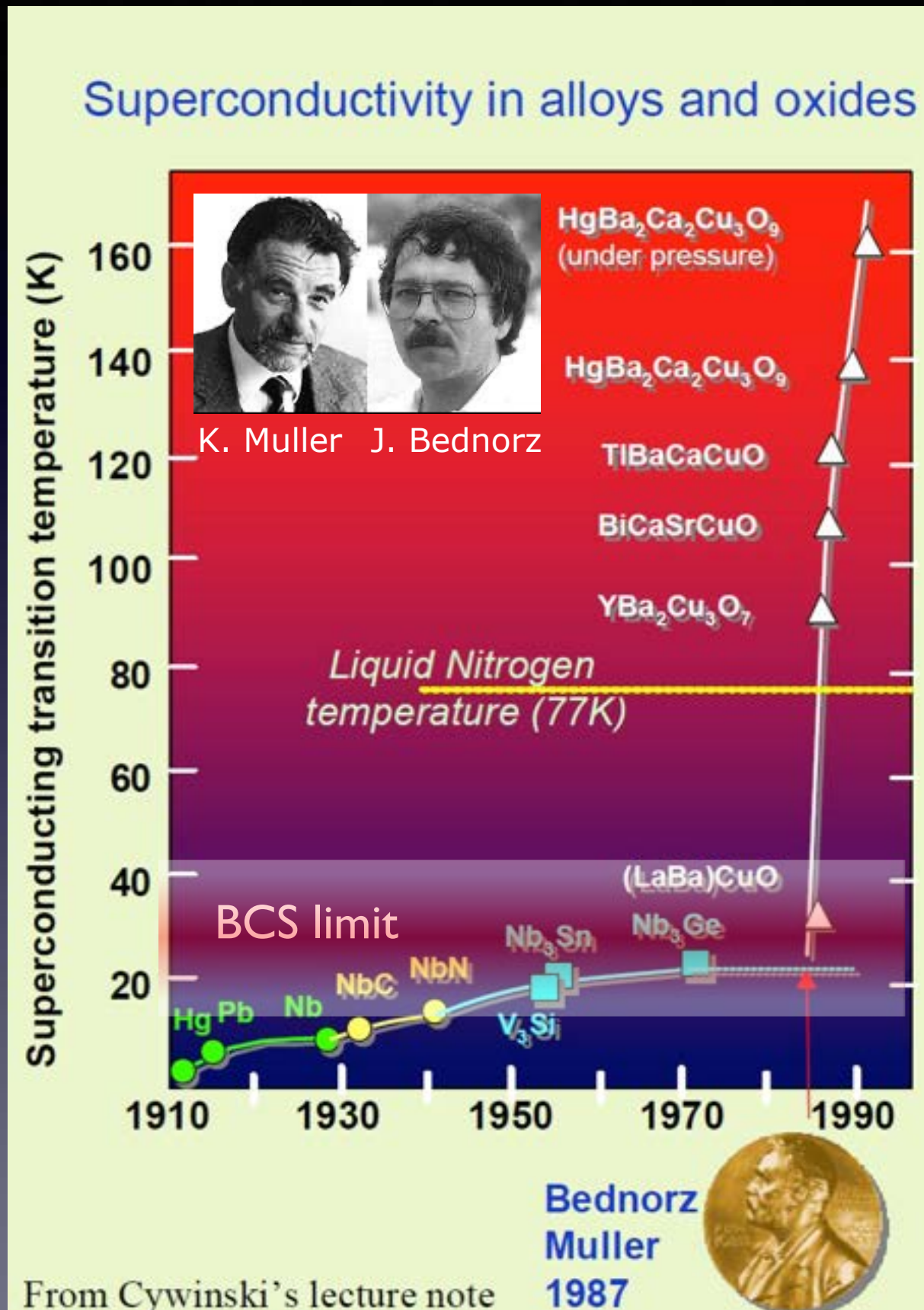


Superconducting Quantum Interference Device (SQUID)

quantum computing

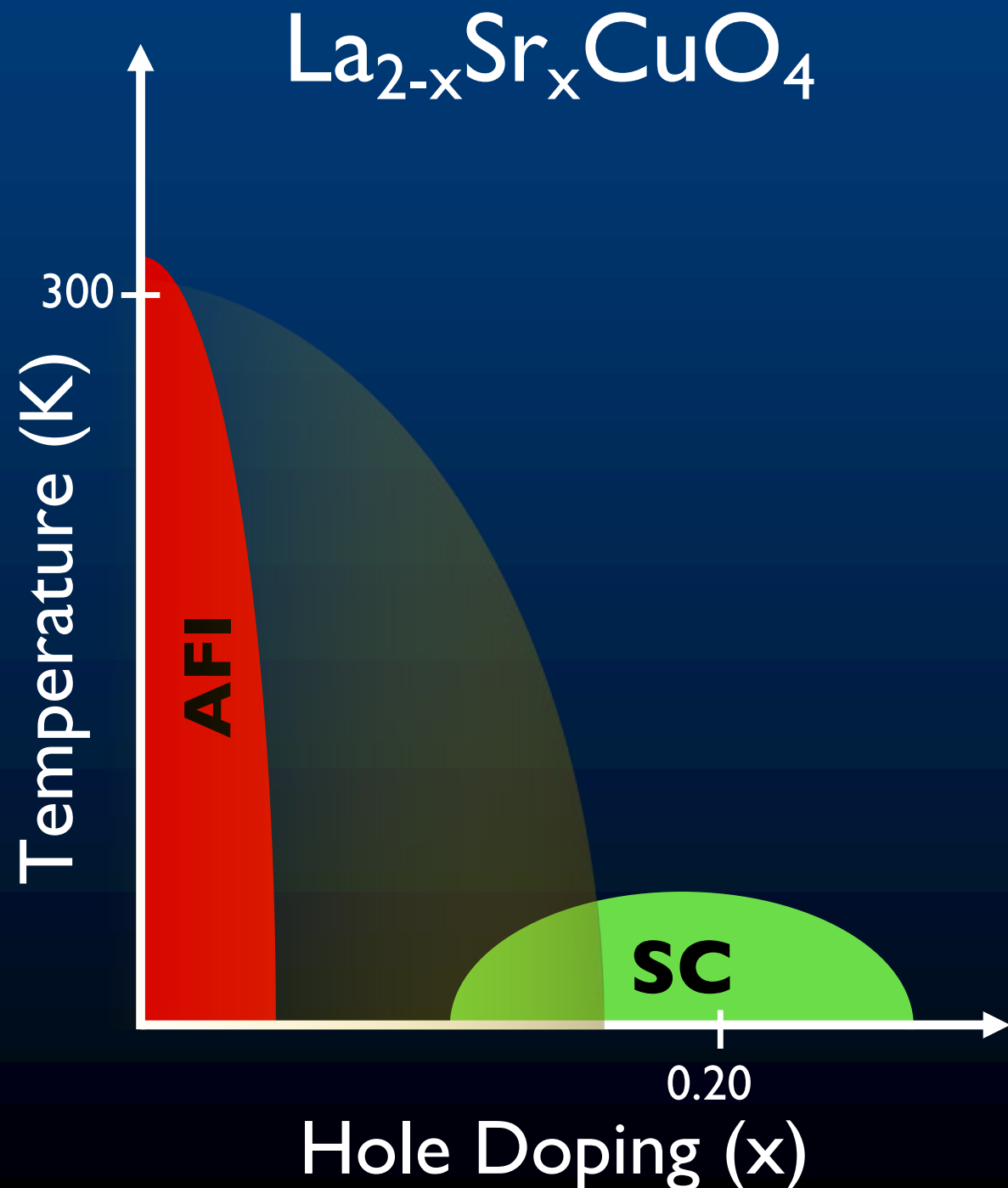


High- T_c Cuprate Superconductors : The “Hydrogen Atom” of Quantum Materials

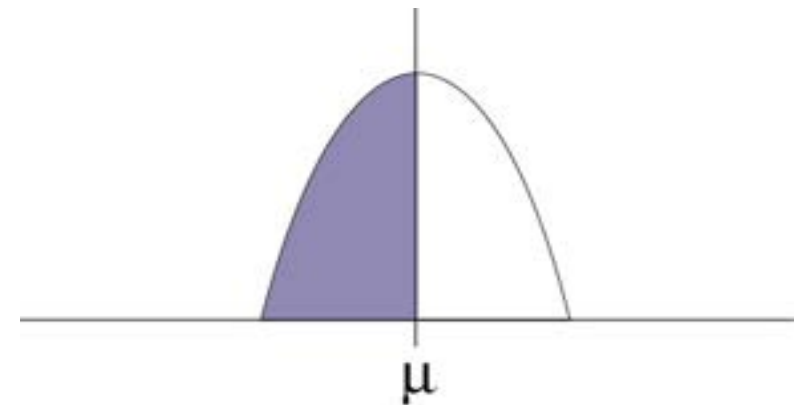
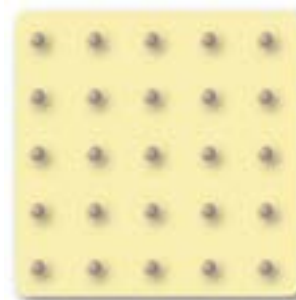


- Conventional “*independent electron*” band theory that works so well for materials like silicon fails completely for cuprates
- Jump-started research on the many-body physics (quantum materials) - “physics of the many”
- Motivated the discovery of many other families of “quantum materials”

Many-Body Interactions in Cuprates : Strong Correlations

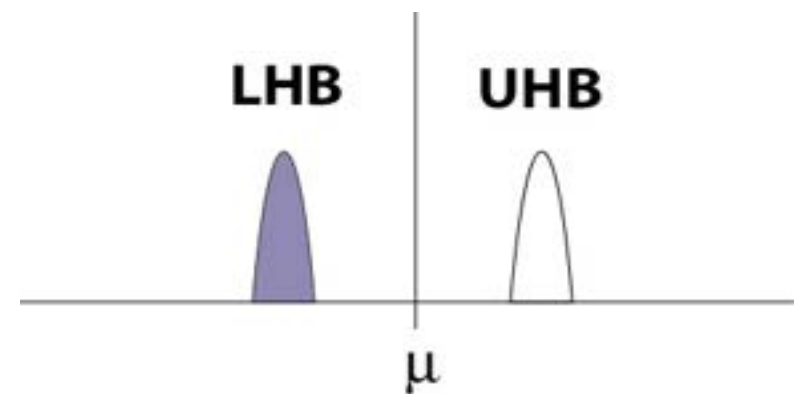


One-Electron Prediction :



Half-filled metal with ~ 3 eV bandwidth

Real Situation :



Mott insulator with ~ 2 eV gap

$$H\psi = E\psi \quad H = \sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 - Ze^2 \sum_R \frac{1}{|ri - R|} \right) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|ri - rj|}$$

independent electrons – indeed unreasonable approximation?

High- T_c is an extension of the long standing problem of insulating oxides – “Mott Insulators”



P.W. Anderson, 1987

$$\mathcal{H} = -t \sum_{i,\sigma} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + h.c. \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

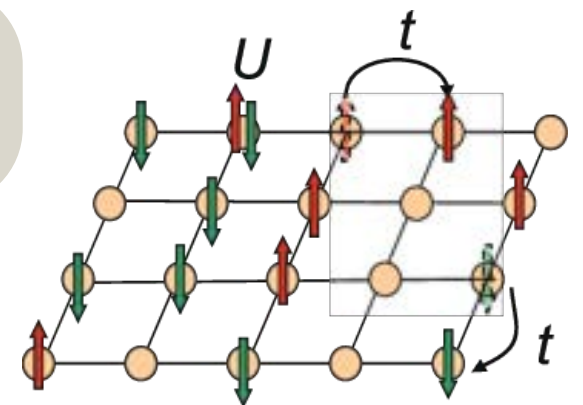
Source of the strong pairing attraction

?

The “hydrogen atom” model of strongly correlated electrons



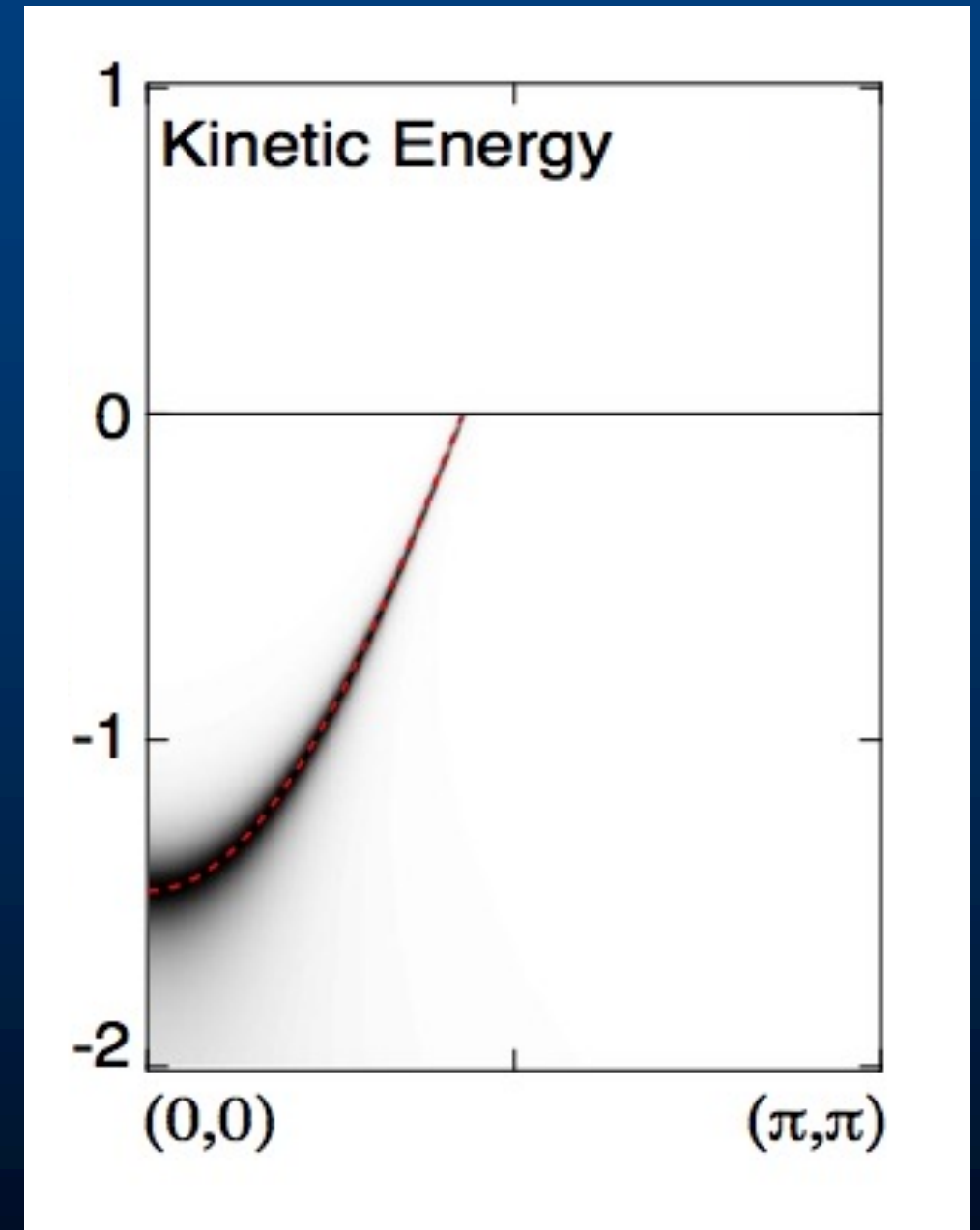
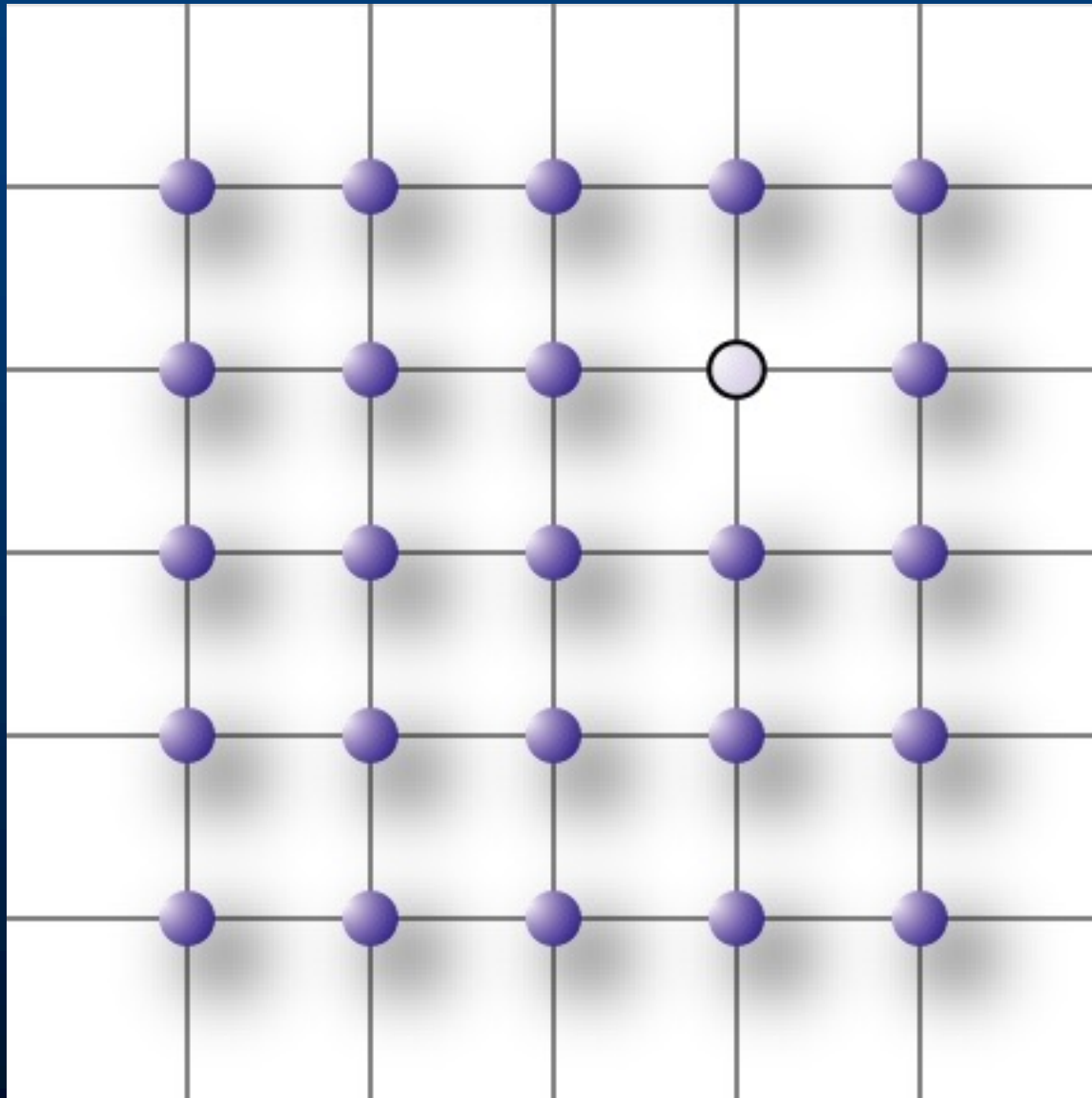
John Hubbard



Courtesy of ZX Shen

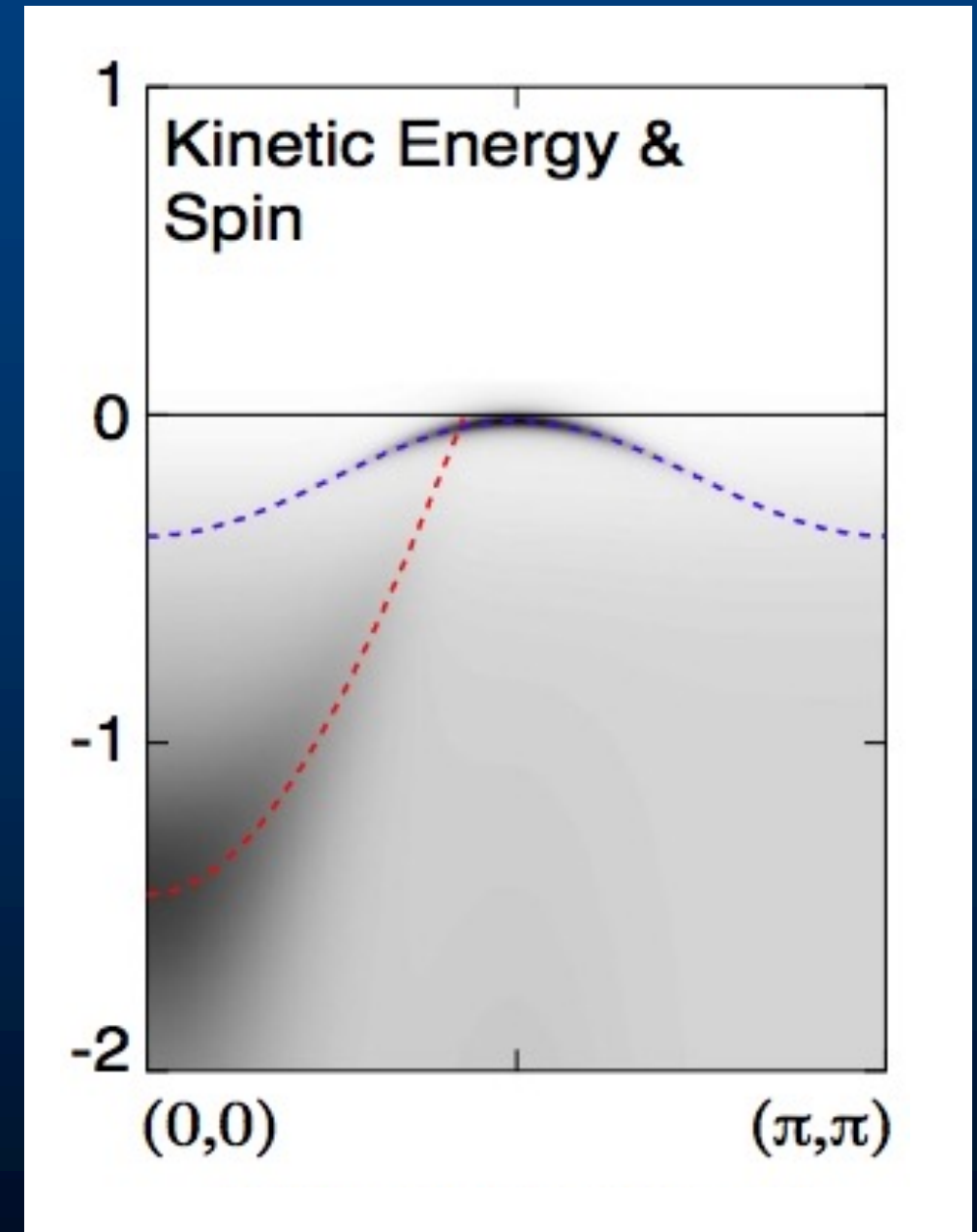
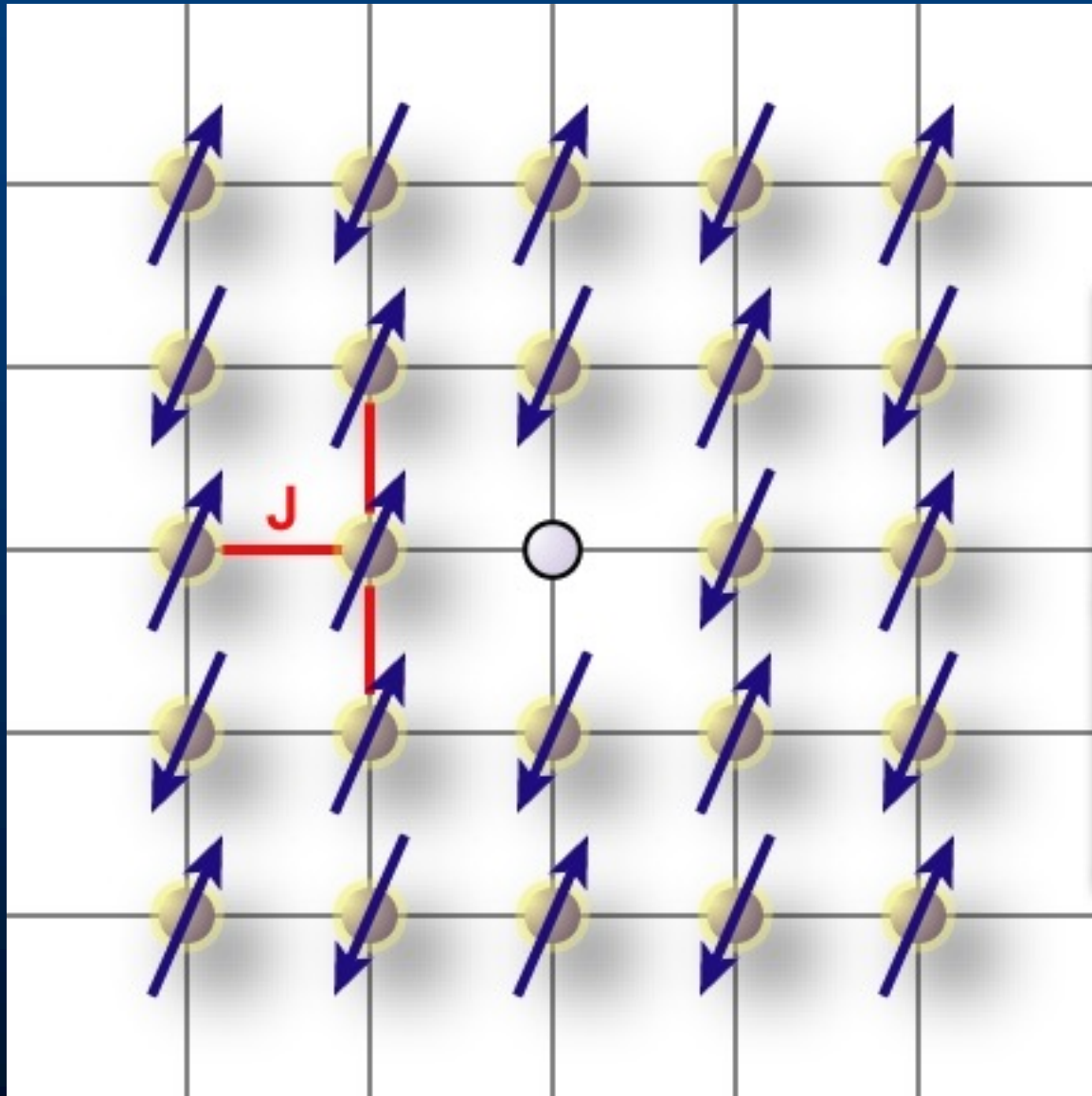
Band Structure Predictions : Non-Interacting

Kinetic Energy Only



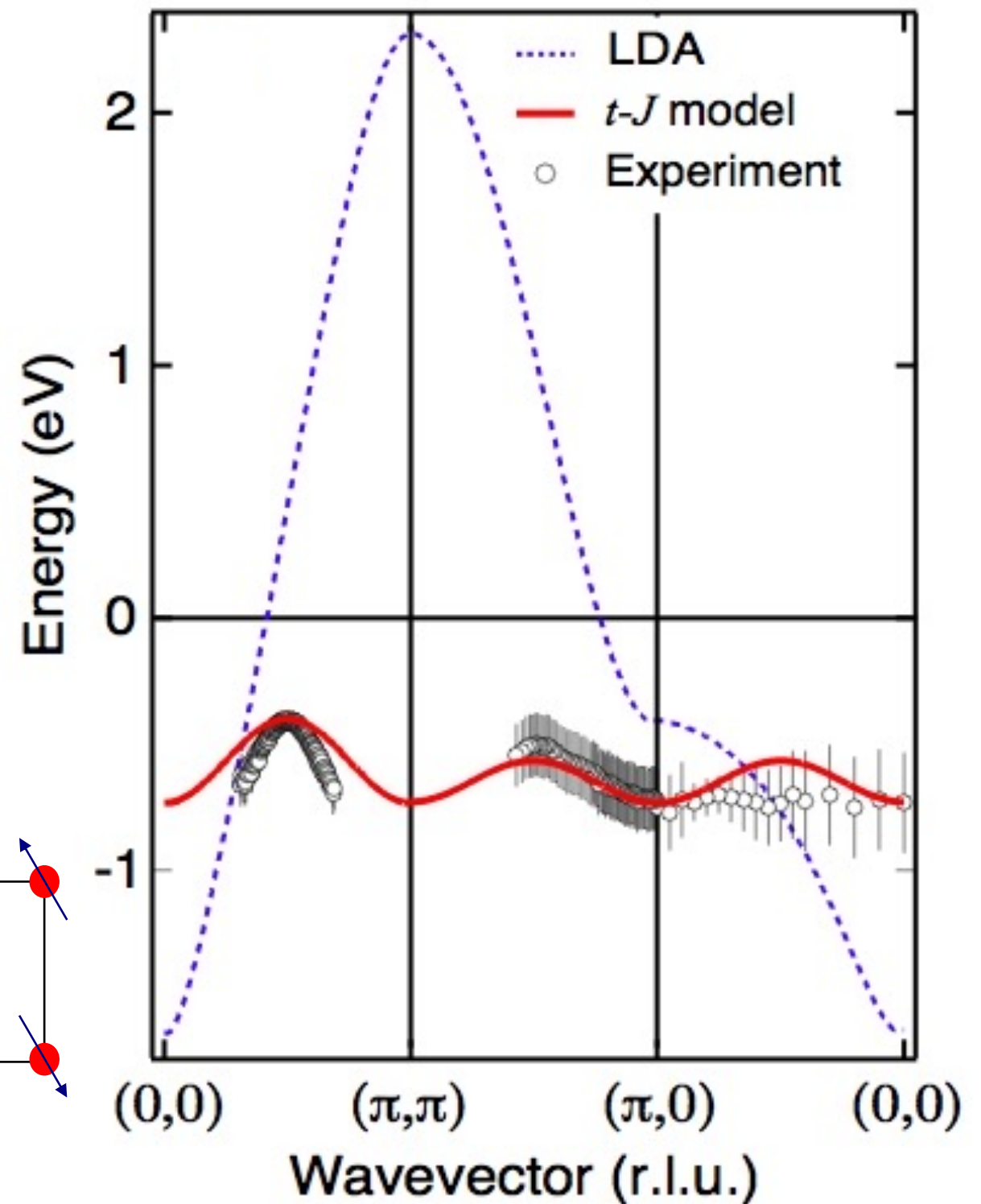
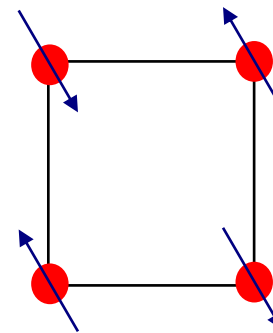
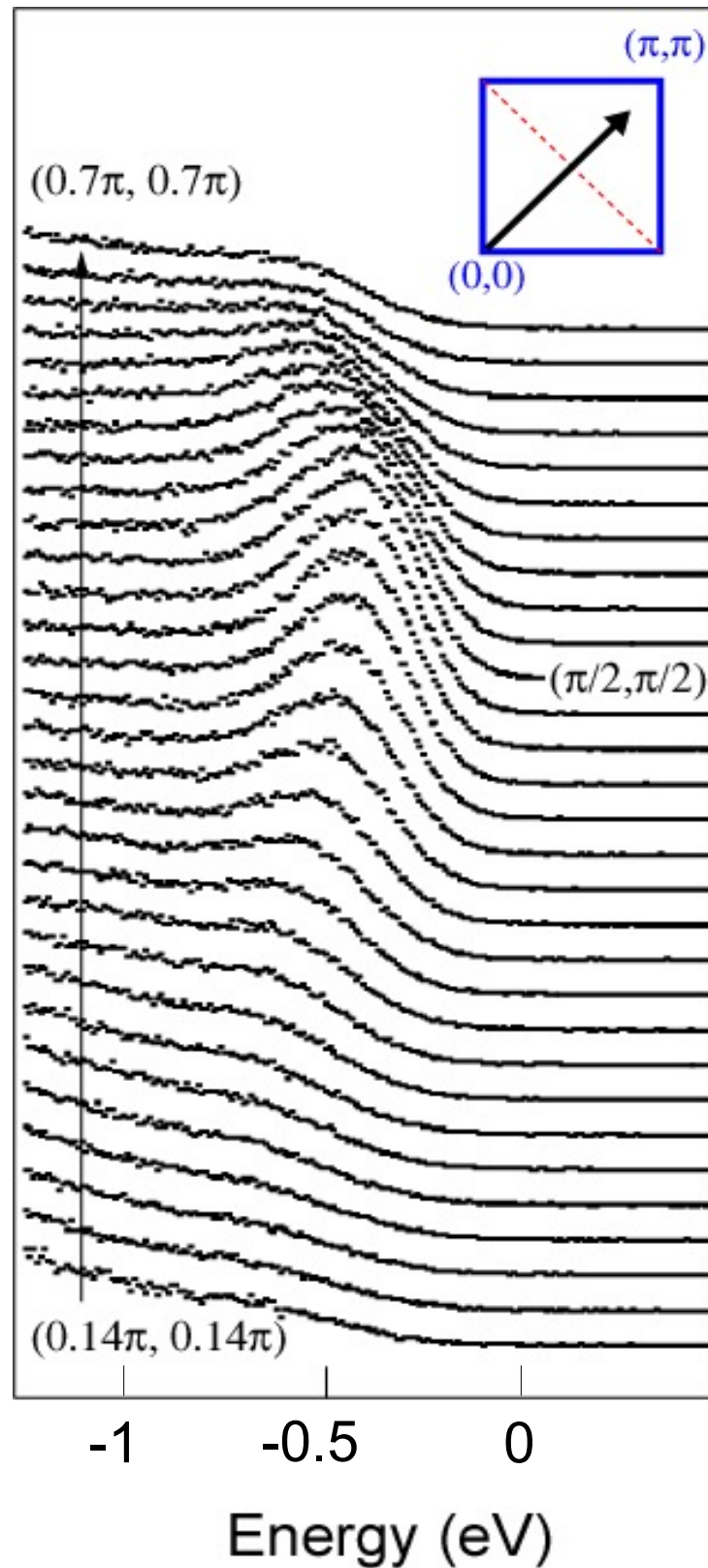
Mott Insulator : Magnetic Interactions

KE & Magnetism

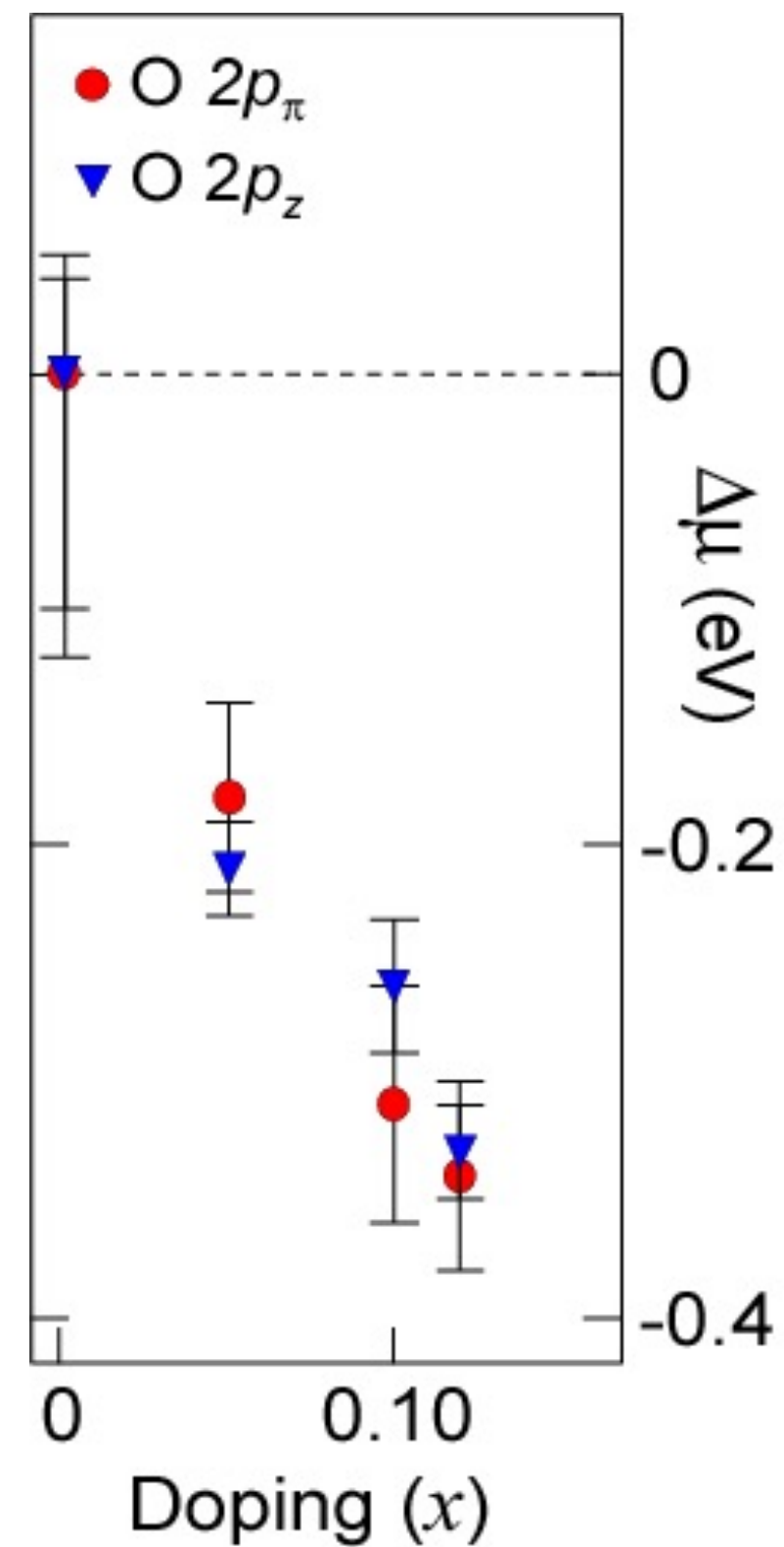
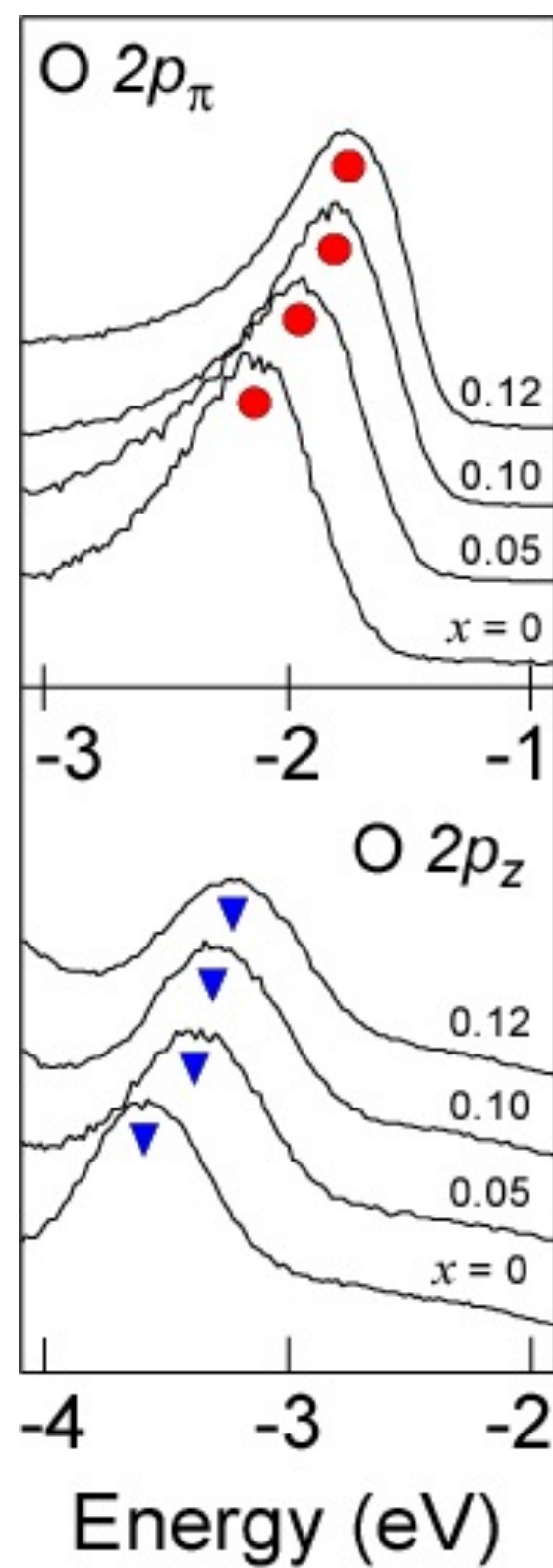
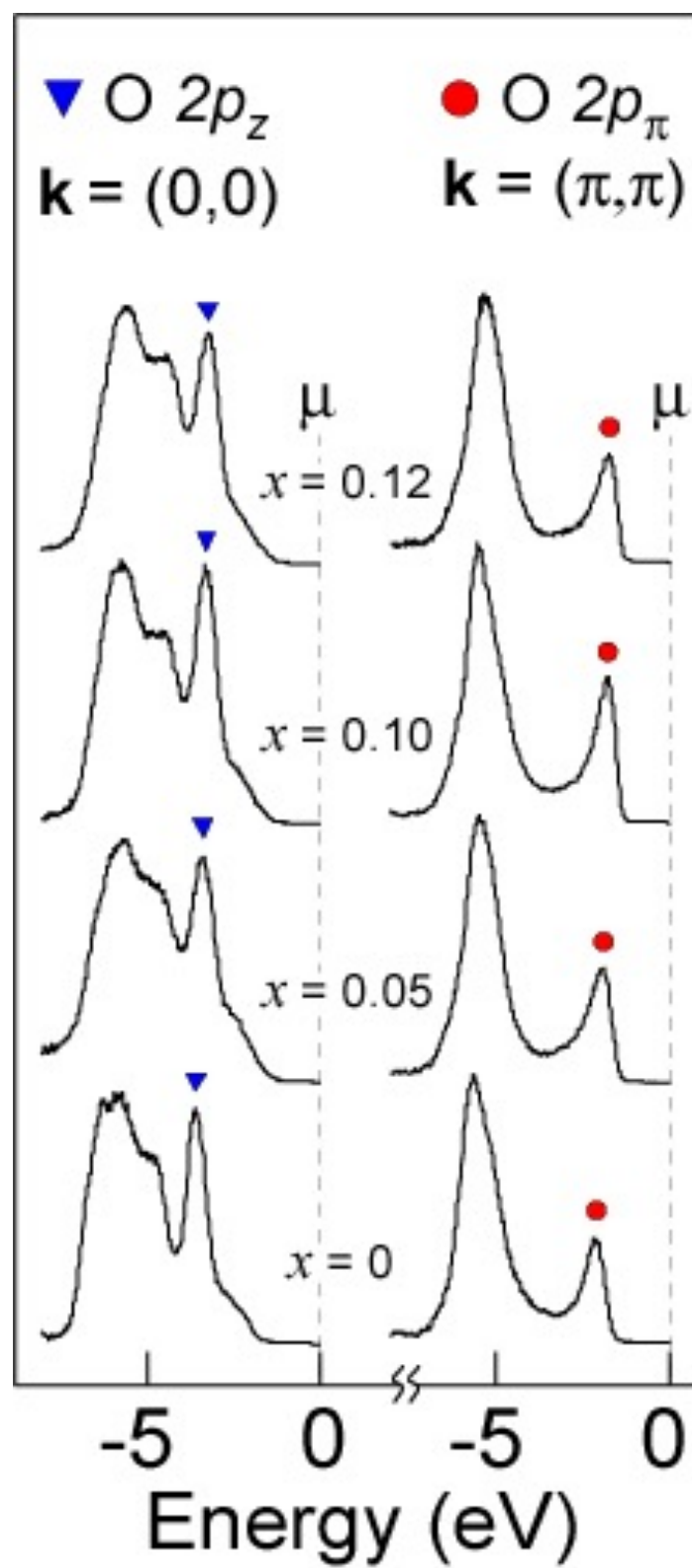


$$t\text{-}J : \quad \mathcal{H} = -t \sum_{\langle ij \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + J \sum_{\langle ij \rangle, \sigma} (\mathbf{S}_i \cdot \mathbf{S}_j - \frac{n_i n_j}{4})$$

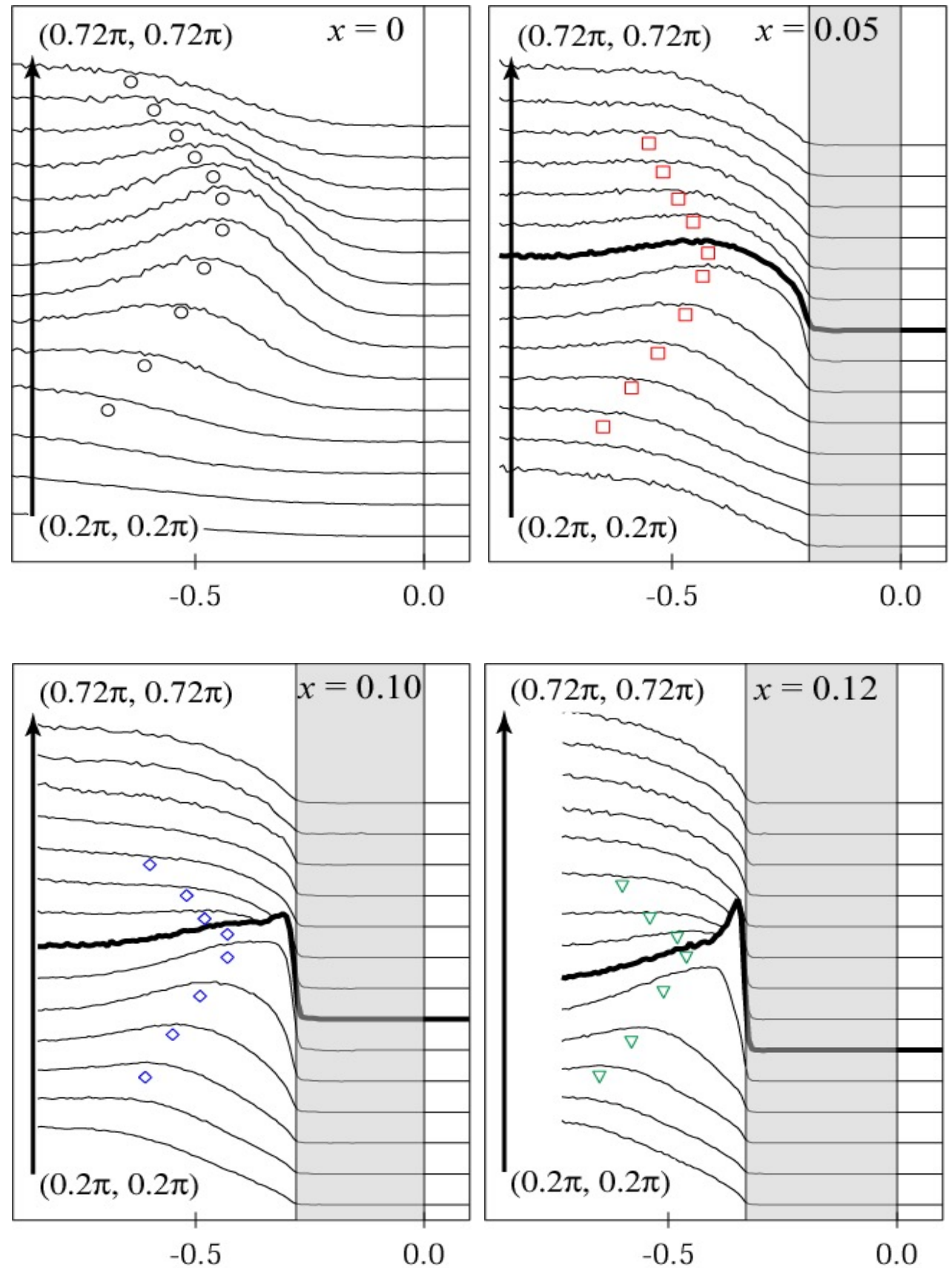
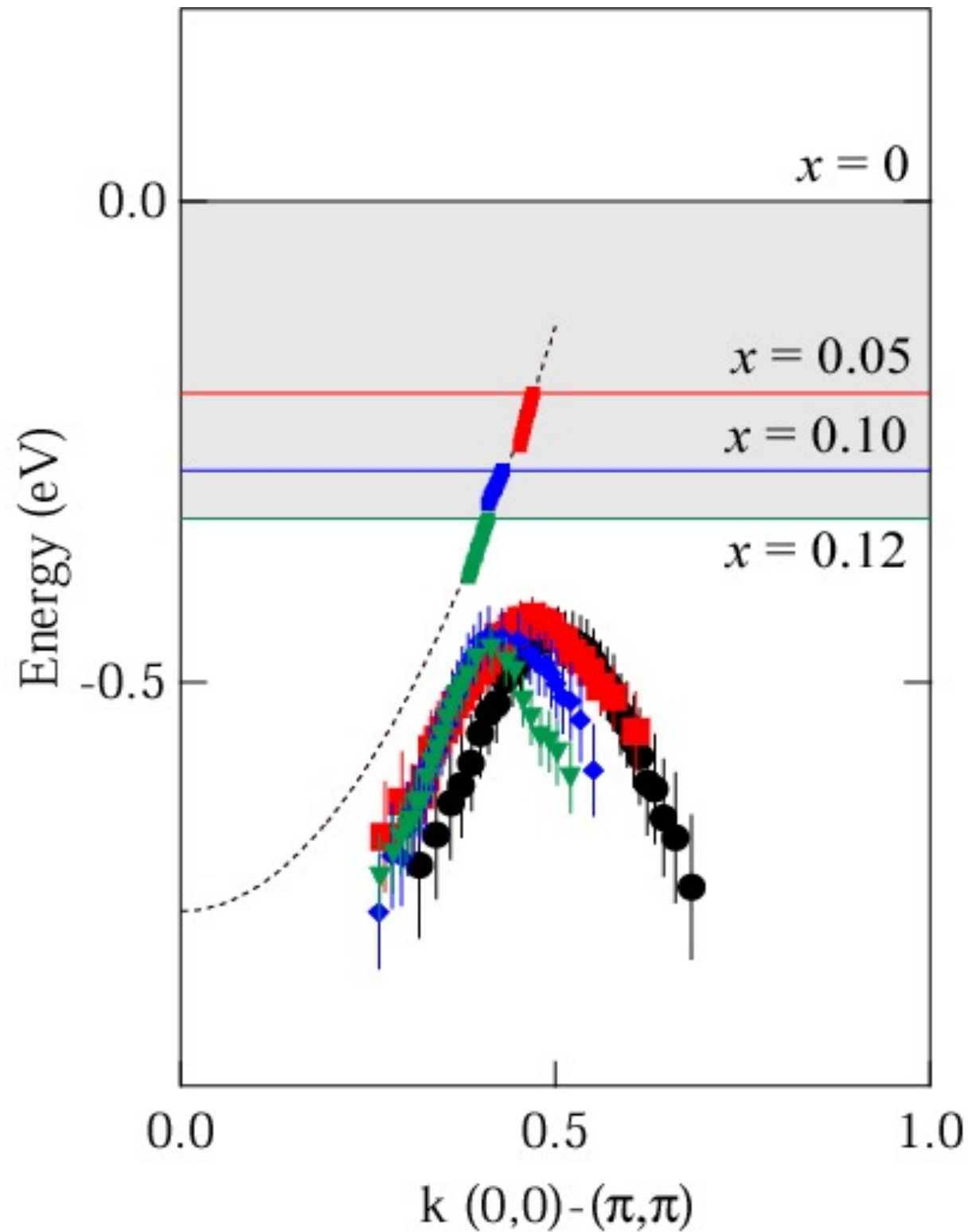
A Single Hole in the Mott Insulator



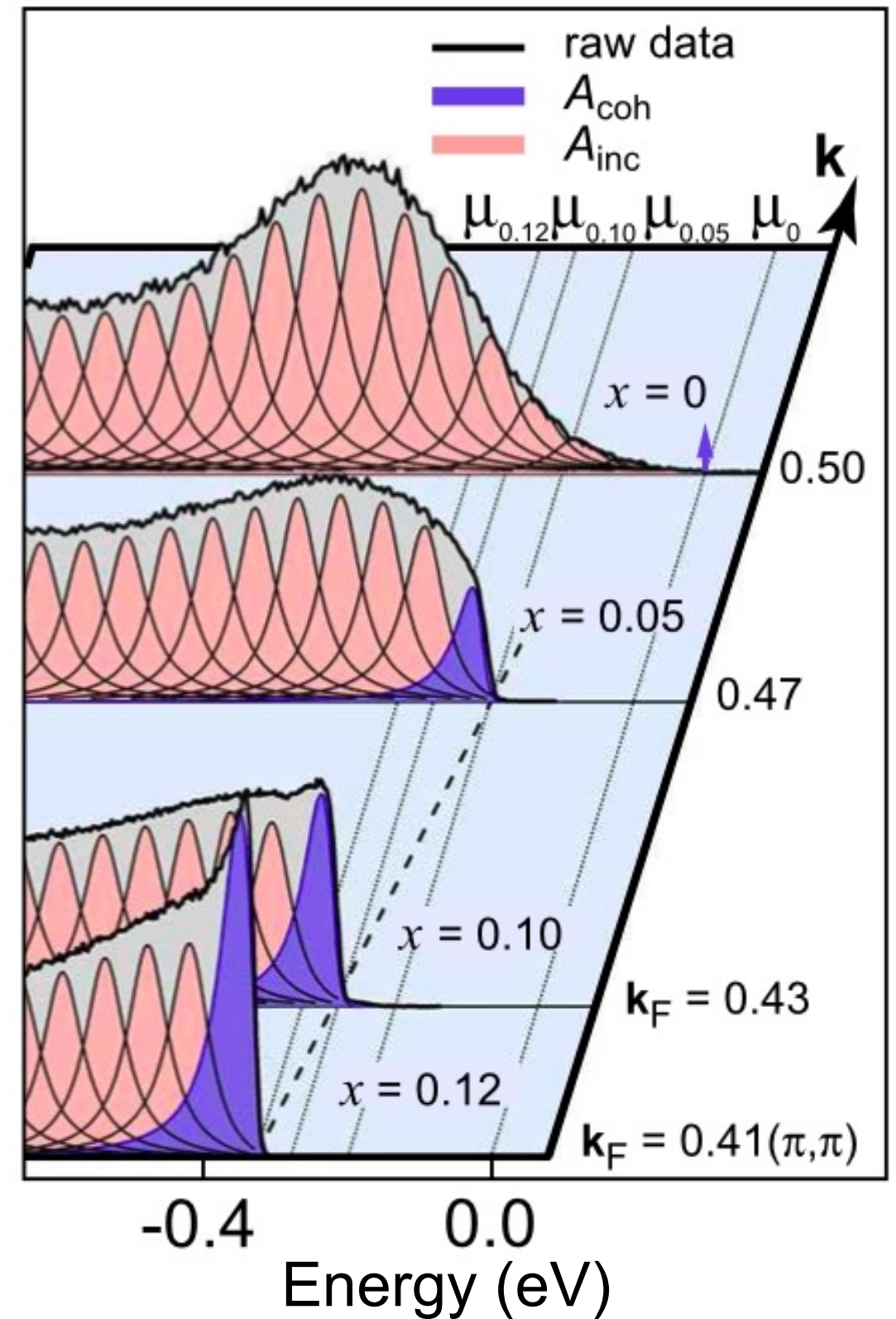
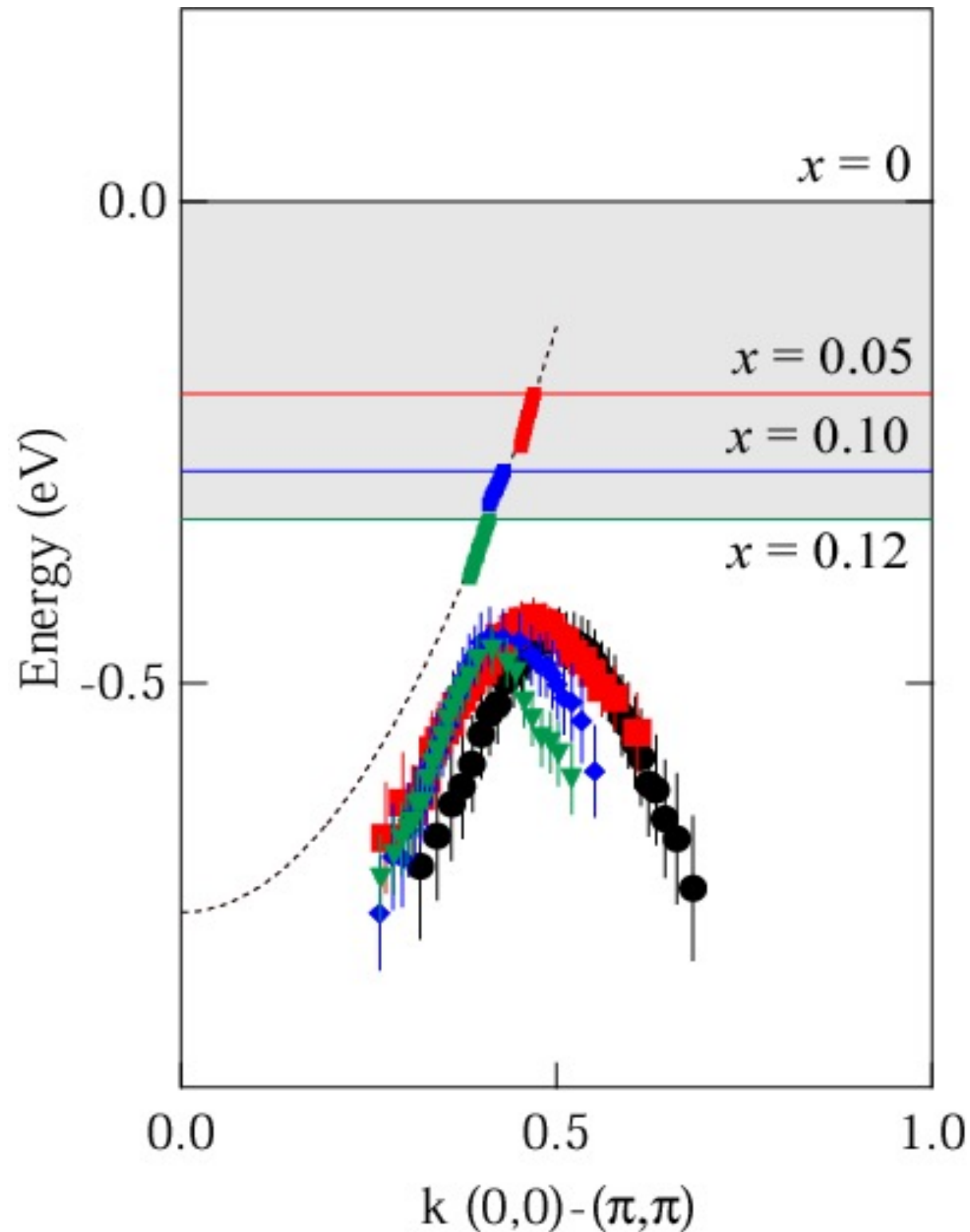
Chemical Potential Shift : $O2p_{\pi}$ & $O2p_z$

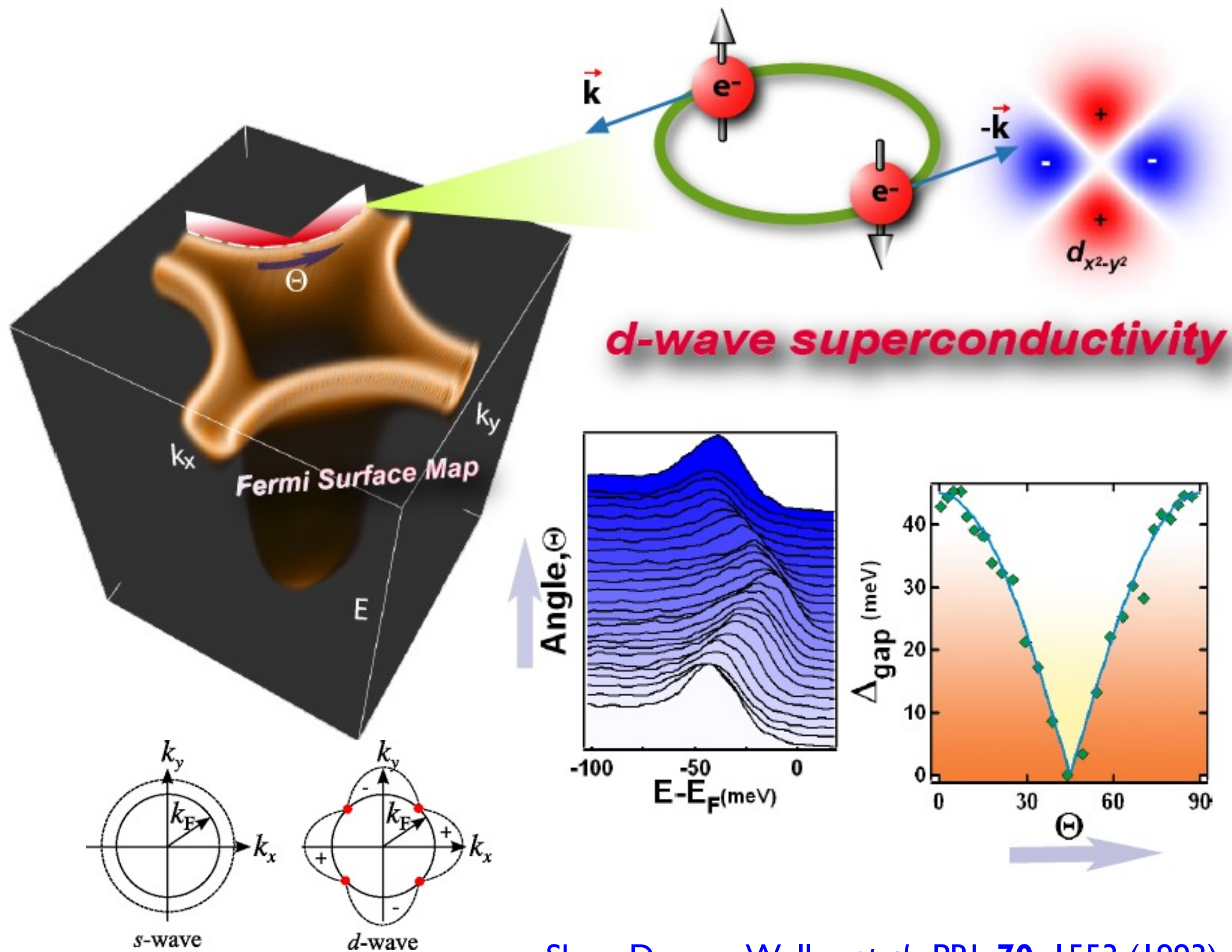


Evolution of Low Energy States with Doping



Evolution of Low Energy States with Doping

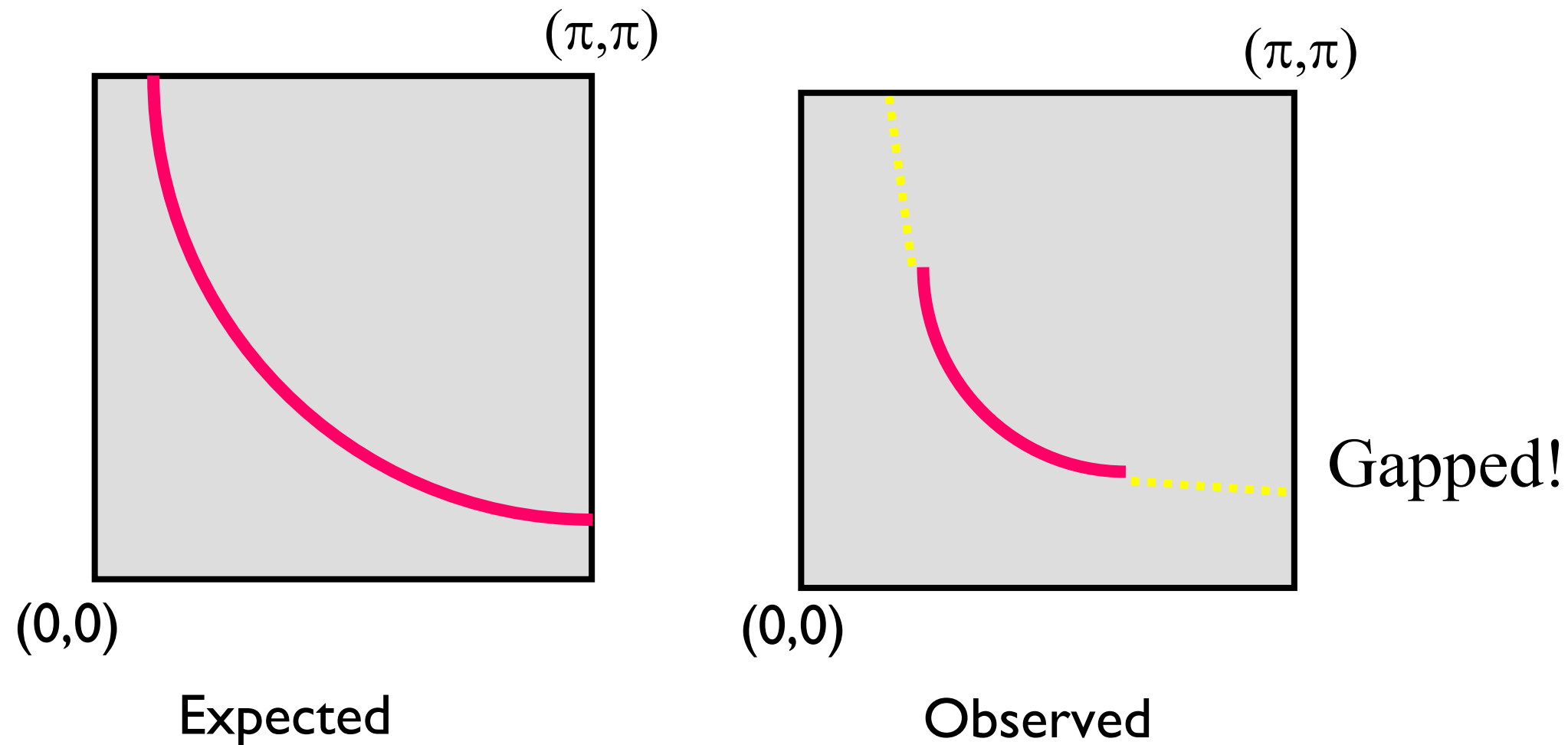




Shen, Dessau, Wells, *et al.*, PRL **70**, 1553 (1993)

P. Bogdanov, Y.L. Chen Ph.D Thesis (2001)

Discontinuous Fermi Surfaces in the Normal State (!!!)

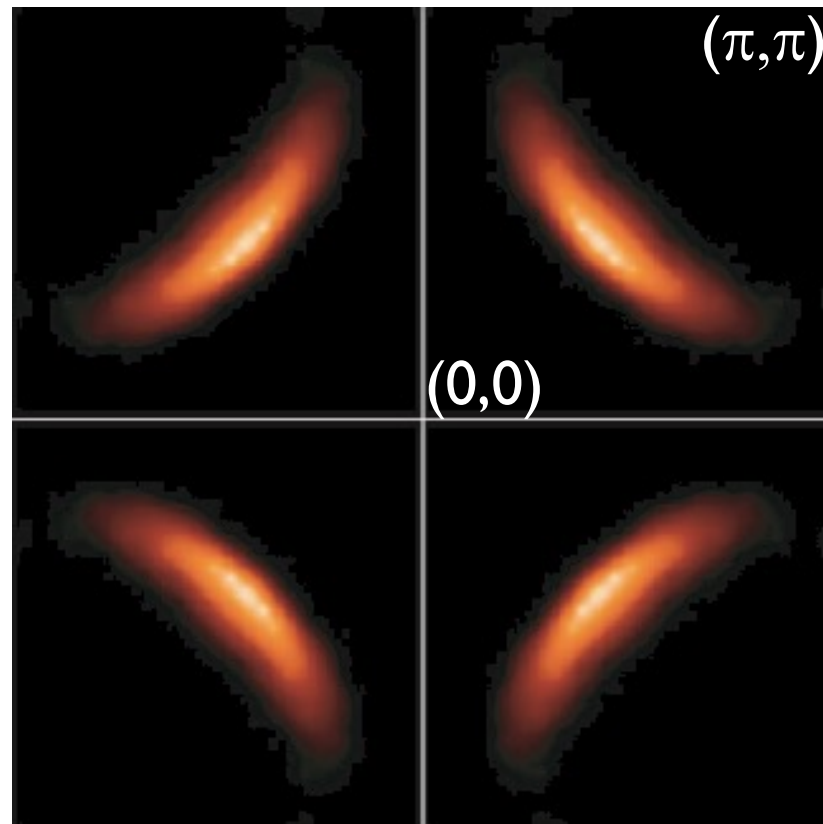


Portion of the Fermi surface gapped, even in the normal state!

D.S. Marshall et al., Phys. Rev. Lett. 76, 4841 (1996)

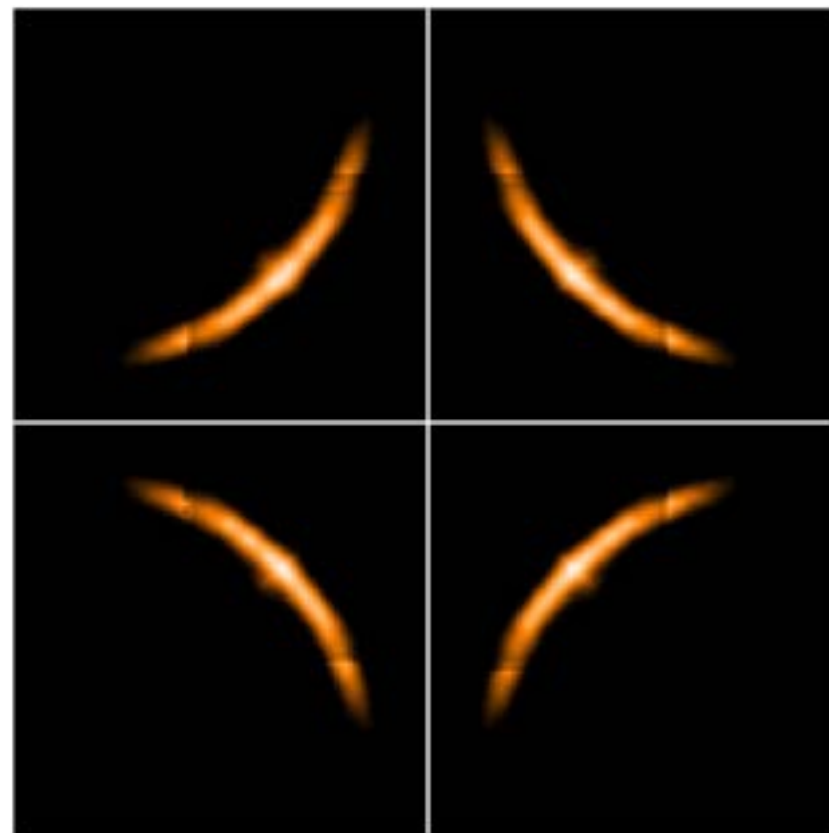
A.G. Loeser et al. Science **273**, 325 (1996)

H. Ding et al. Nature **382**, 51 (1996)



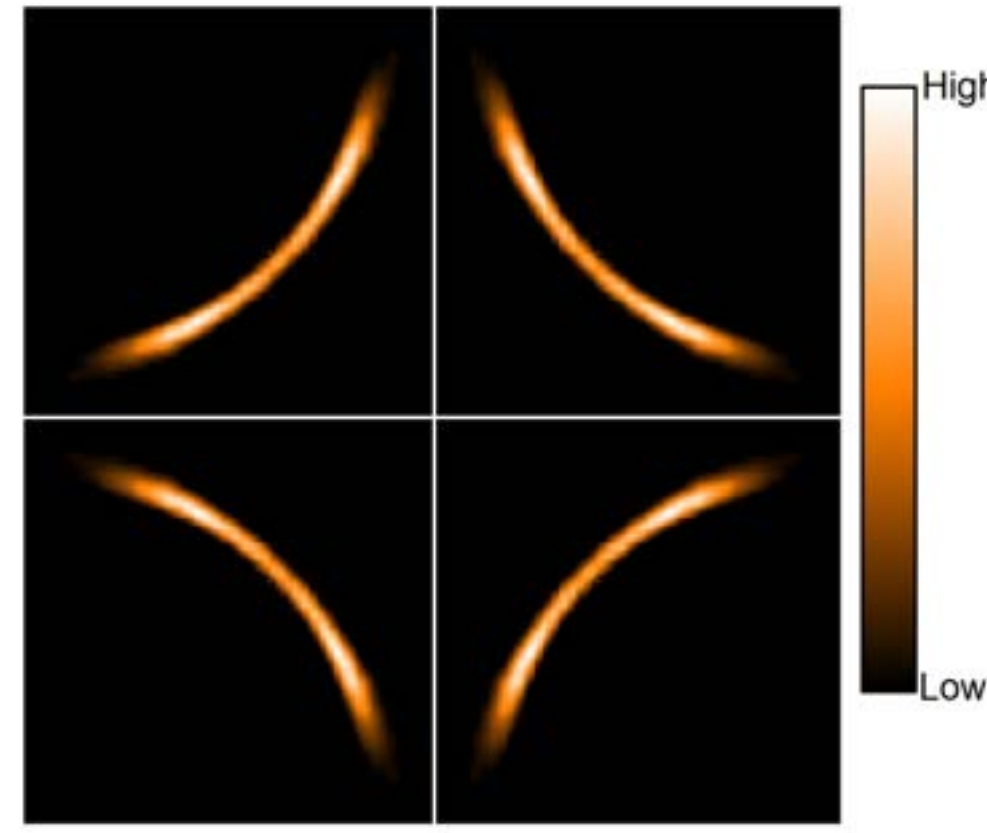
CCOC

K.M. Shen et al.,
Science 307, 901



Bi-2212

W. S. Lee *et al.*
Nature 450, 81



Bi-2201

M. Hashimoto et al.,
Nature Physics 6, 414

Example #1 : High-Temperature Cuprate Superconductors

- Evolution from the parent Mott insulating state
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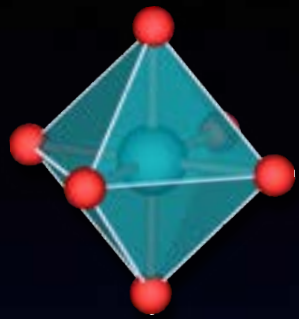
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Ruthenate properties are highly tunable with structural changes

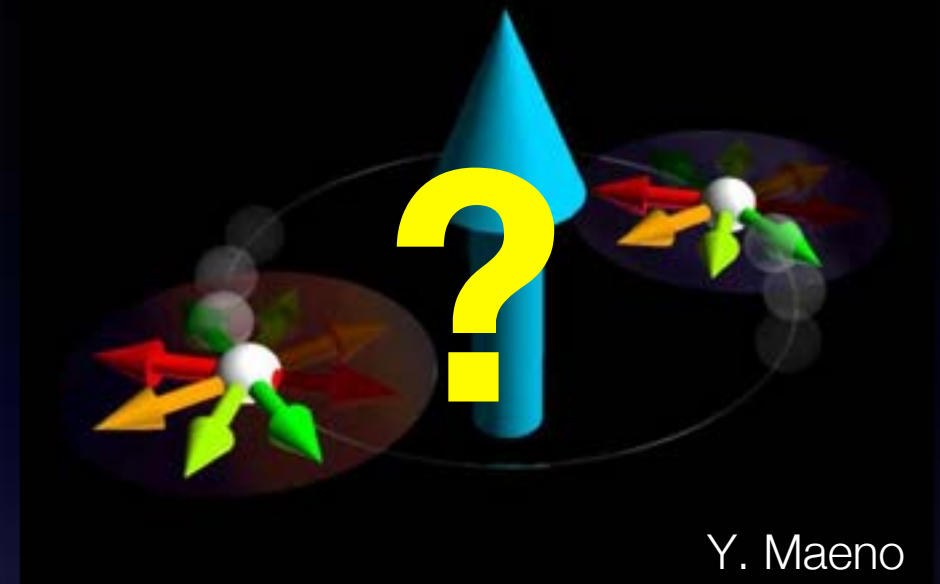


RuO_6 octahedra

$\text{Ru}^{4+} : 4d^4$

Compound	Dimensionality	Octahedral Connectivity	Properties
Sr_2RuO_4	2D	CORNER	Exotic SC
Ca_2RuO_4	2D	CORNER	AF Mott Insulator
CaRuO_3	3D	CORNER	heavy FL
SrRuO_3	3D	CORNER	FM Metal
RuO_2	3D	EDGE & CORNER	Metal

ground states can be tuned from metal, AF insulator, FM metal, exotic SC, simply by changing connectivity of RuO_6 octahedra (without doping)

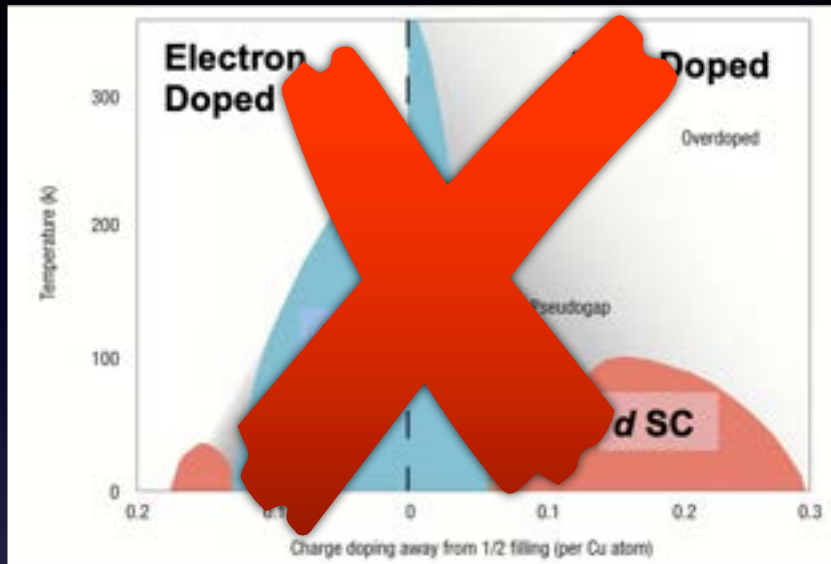


- various experiments (μSR , Kerr rotation) point towards broken time-reversal symmetry
- simple chiral p -wave, spin-triplet model called into question by recent experiments
- order parameter is unconventional, but precise nature still up for debate

WANTED : clean knobs to control SC in Sr_2RuO_4

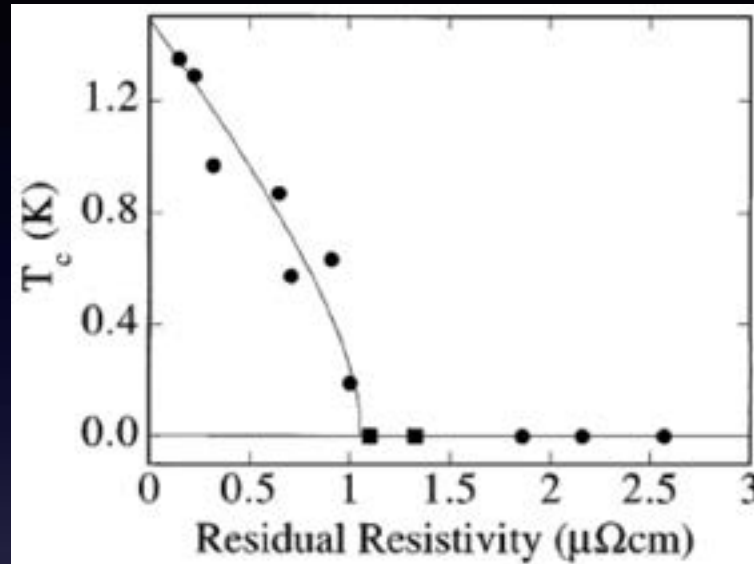
traditional approaches like doping and chemical substitution cannot be applied to studying superconductivity of Sr_2RuO_4

uniaxial or epitaxial (biaxial) strain is a clean alternative



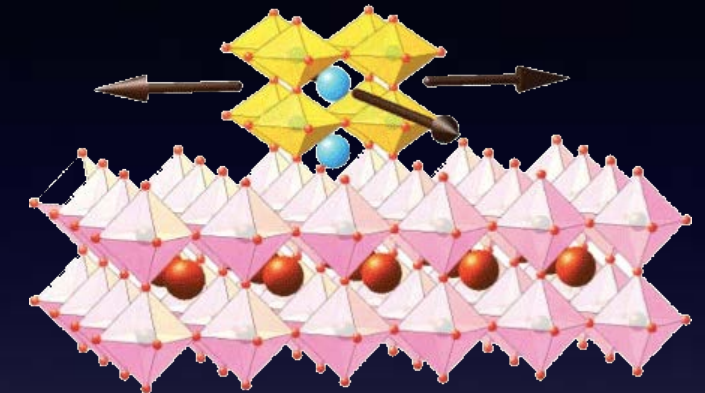
D.A. Bonn, Nature Phys. (2007)

- HTSCs (cuprates, Fe-SC) require doping at the level of 10% to realize SC
- SC is robust at the level of 100,000s of ppm's!
- superconducting coherence lengths ~ 1 nm



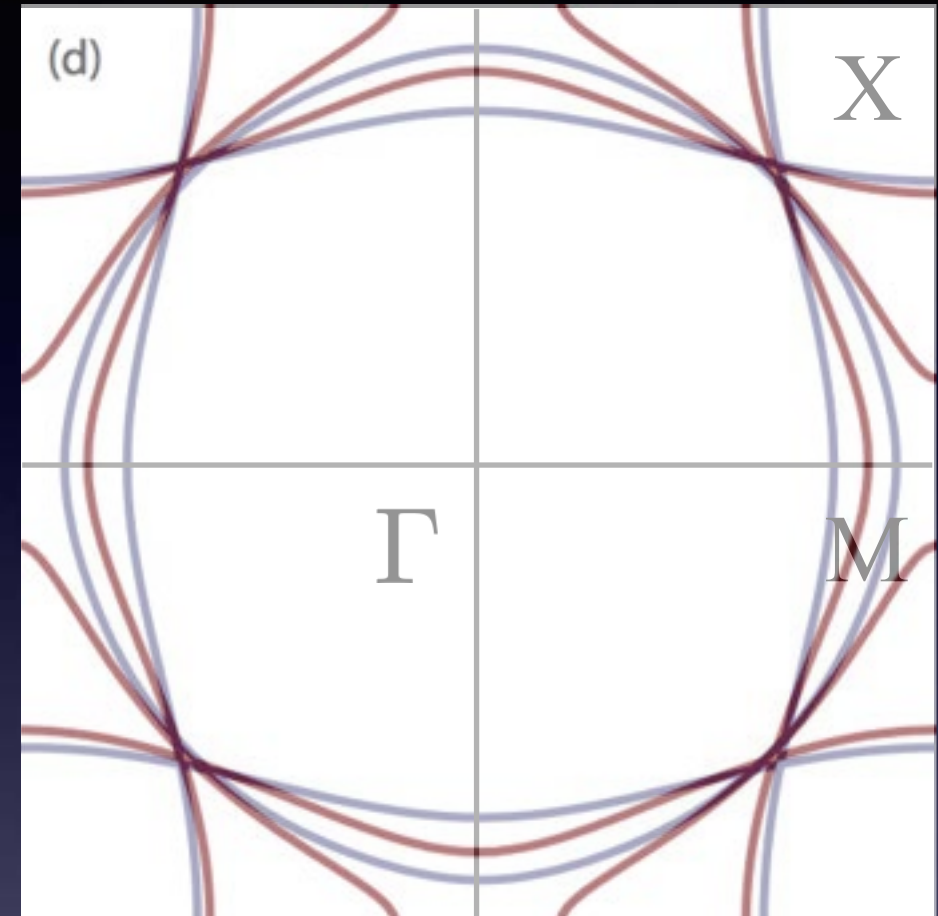
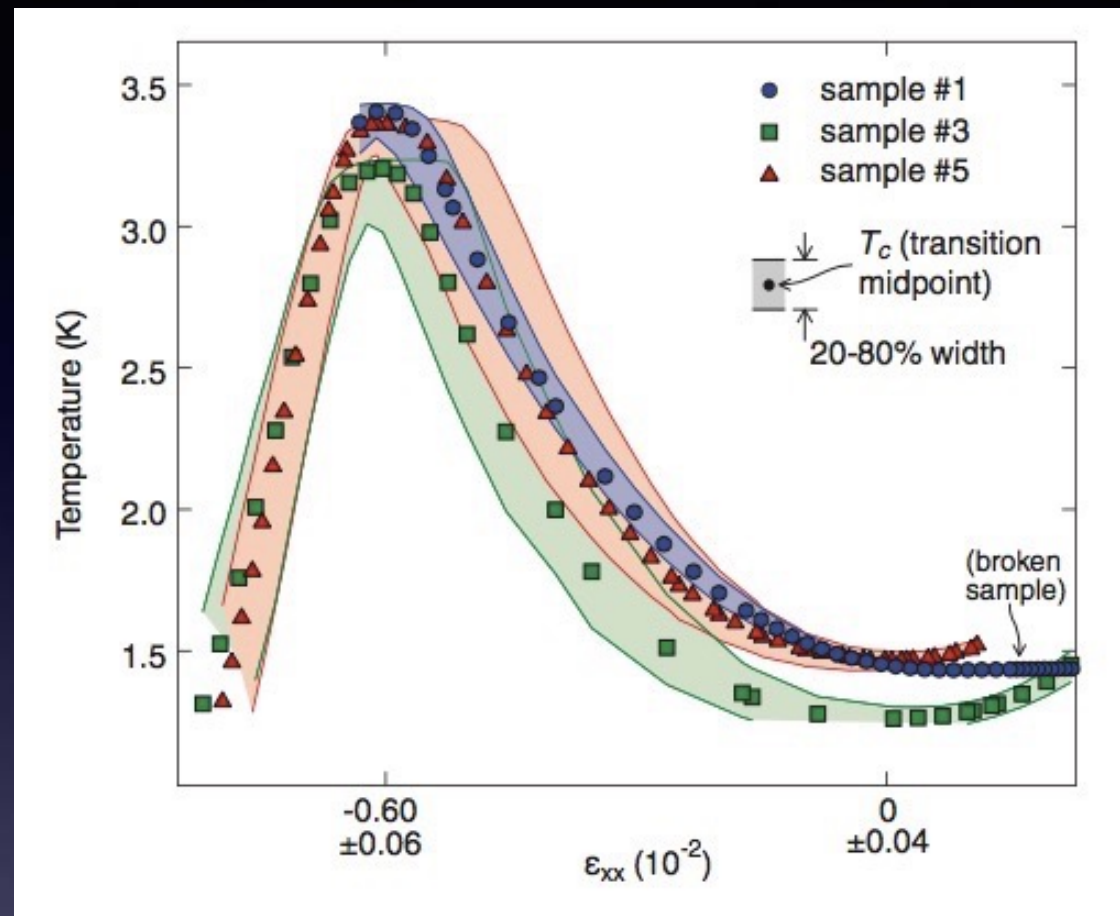
A.P. Mackenzie et. al, *Phys. Rev. Lett.* **80**, 161 (1998)

- Sr_2RuO_4 is the most disorder-sensitive SC known
- tens to hundreds of ppm's of impurities kills SC
- superconducting coherence lengths ~ 0.1 microns



- strains on the order of a couple percent can be applied
- does not introduce substantial disorder
- can also be implemented in device structures

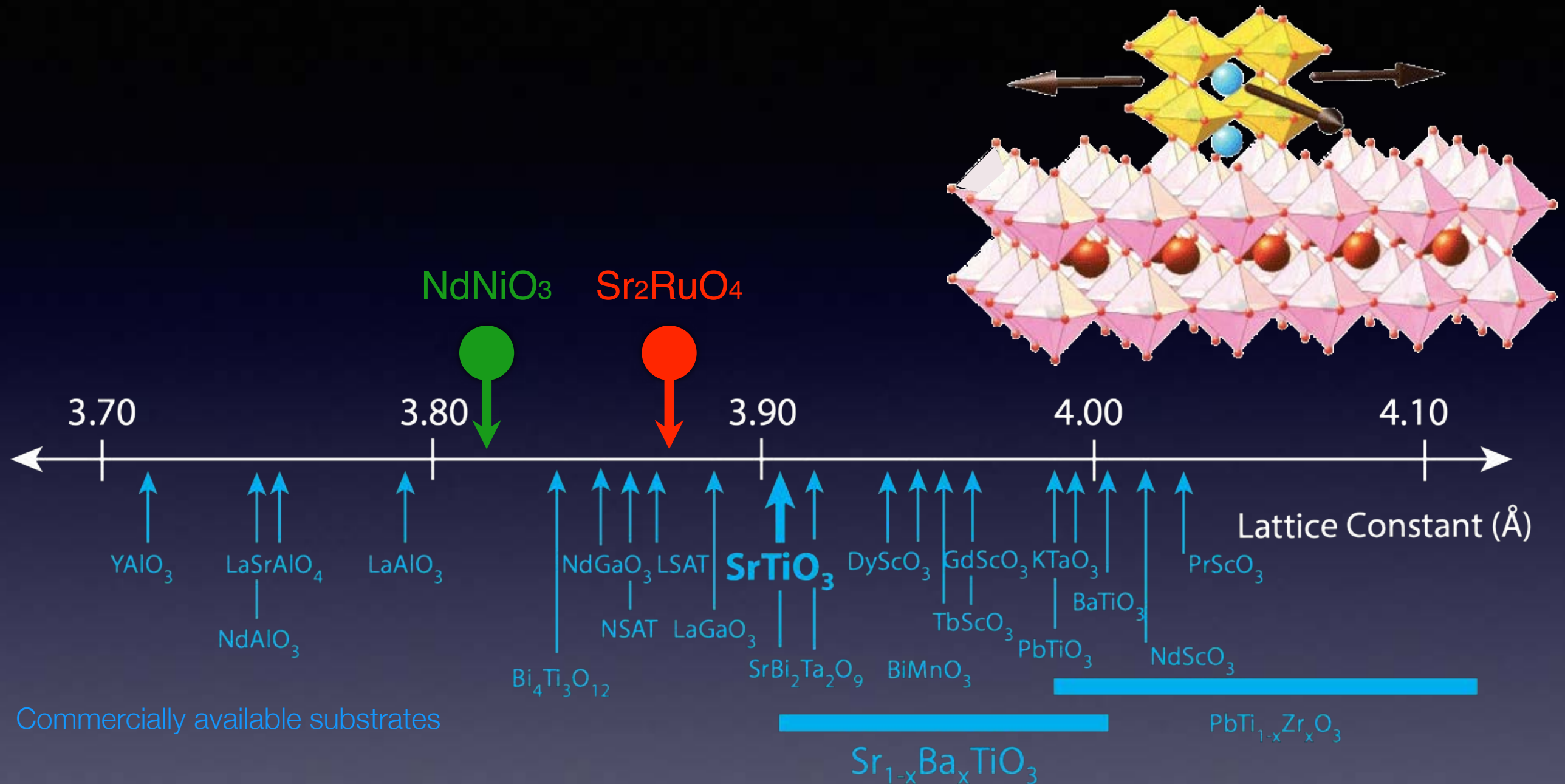
in-plane uniaxial strain significantly increases T_c in Sr_2RuO_4



enhancements in T_c may be tied to proximity of van Hove singularity to E_F ; proposed that "Lifshitz transition" likely gives rise to the sharp peak in T_c with strain.

How does electronic structure evolve with epitaxial strain?

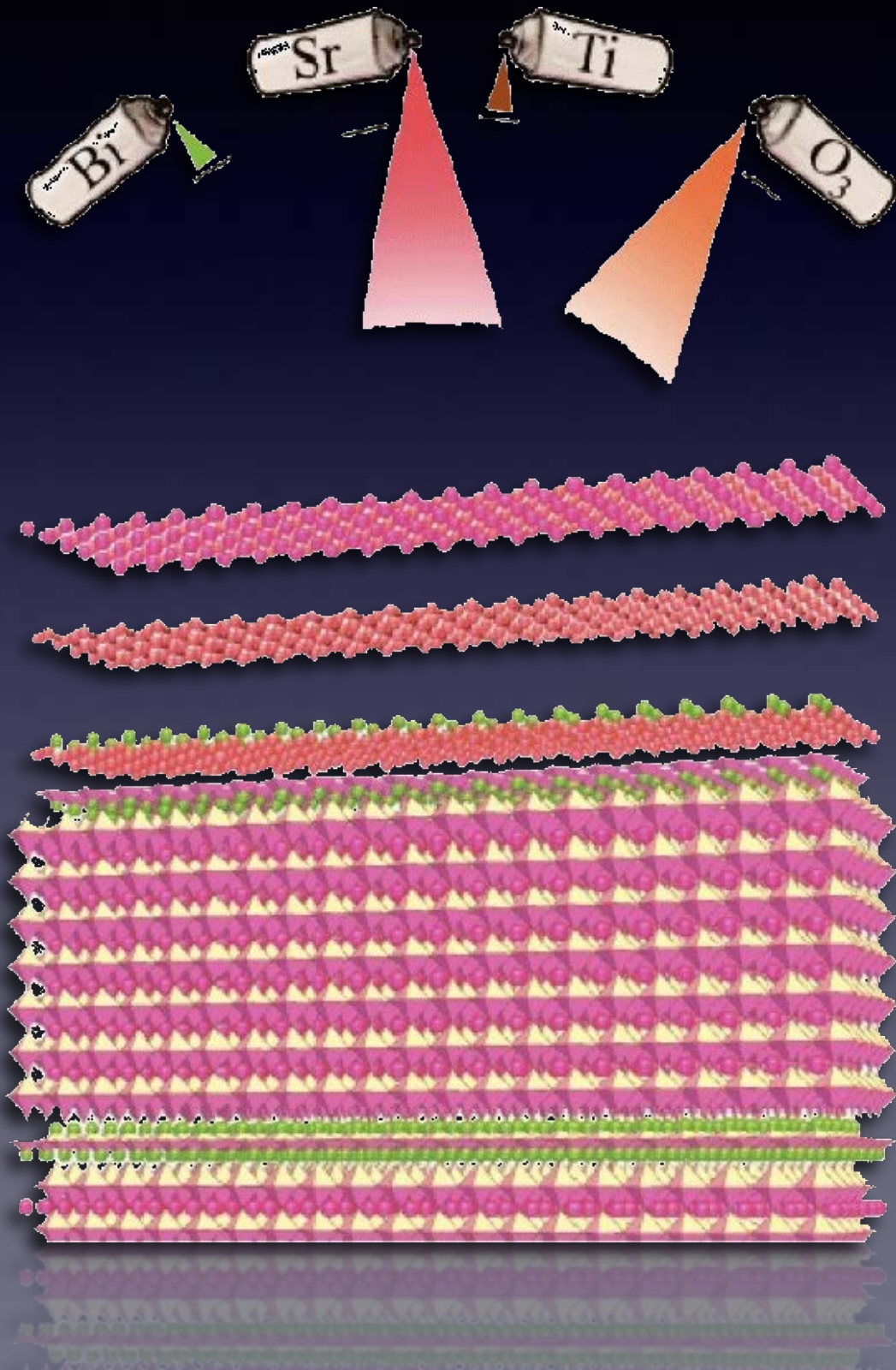
epitaxial strain as a tuning parameter in quantum material heterostructures



- clean tuning parameter (unlike chemical pressure)
- enables most spectroscopies & probes (unlike hydrostatic pressure)
- much larger strains than possible in bulk crystals (and different symmetries), ~3%
- scalable and enables device fabrication (e.g. strained silicon MOSFETs)

molecular beam epitaxy (MBE)

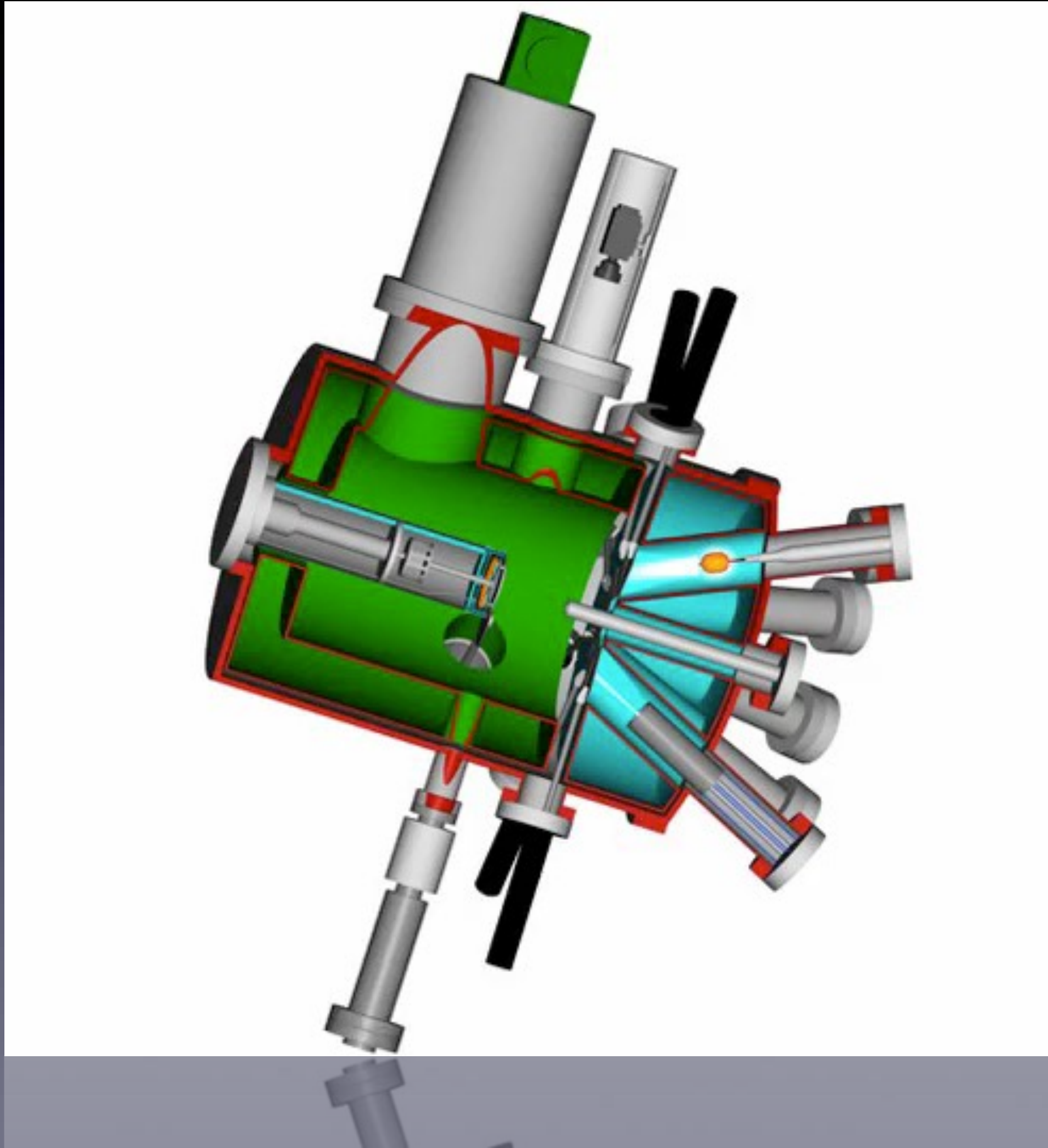
*“atomic spray
painting”*



advantages of MBE

- sub-monolayer control of atomic layers
- can create nearly perfect atomic interfaces, heterostructures, or metastable structures not possible in bulk
- can synthesize materials of extremely high purity
- used for synthesizing laser diodes, LEDs, photovoltaics, etc....

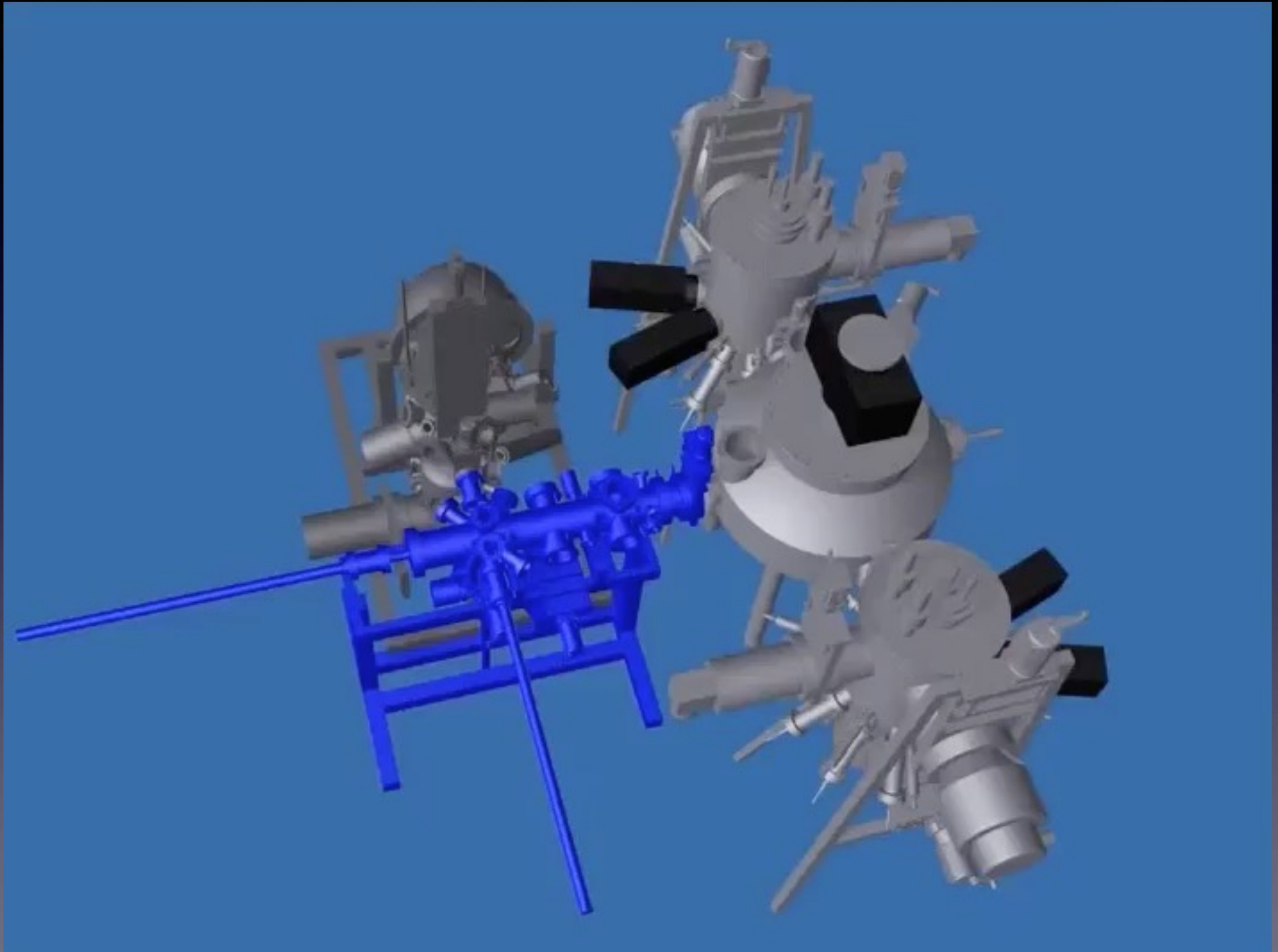
molecular beam epitaxy (MBE)



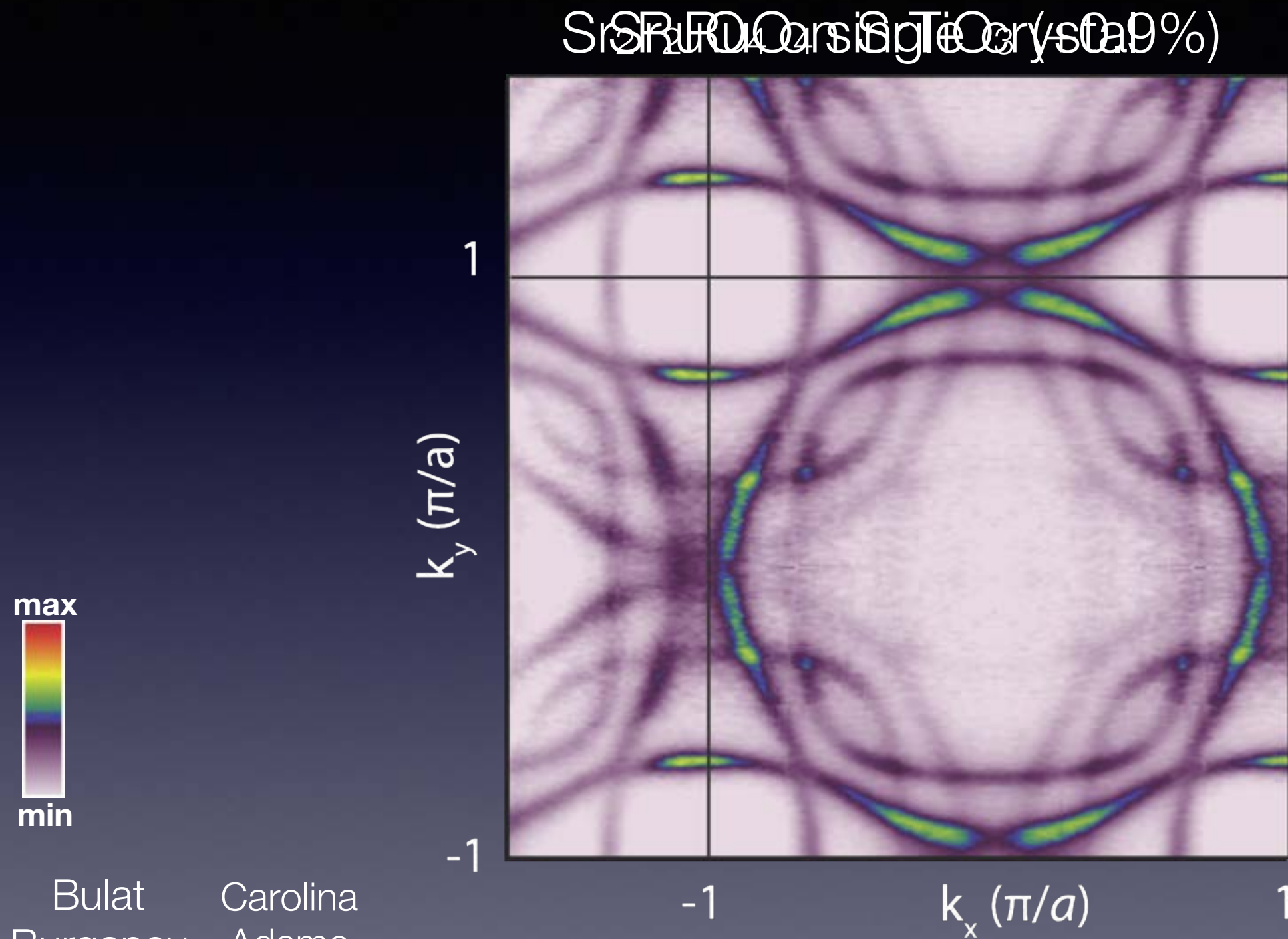
integrated ARPES & MBE system



Darrell
Schlom



Can tensile strain push the van Hove singularity closer to E_F ?



Bulat
Burganov

Carolina
Adamo



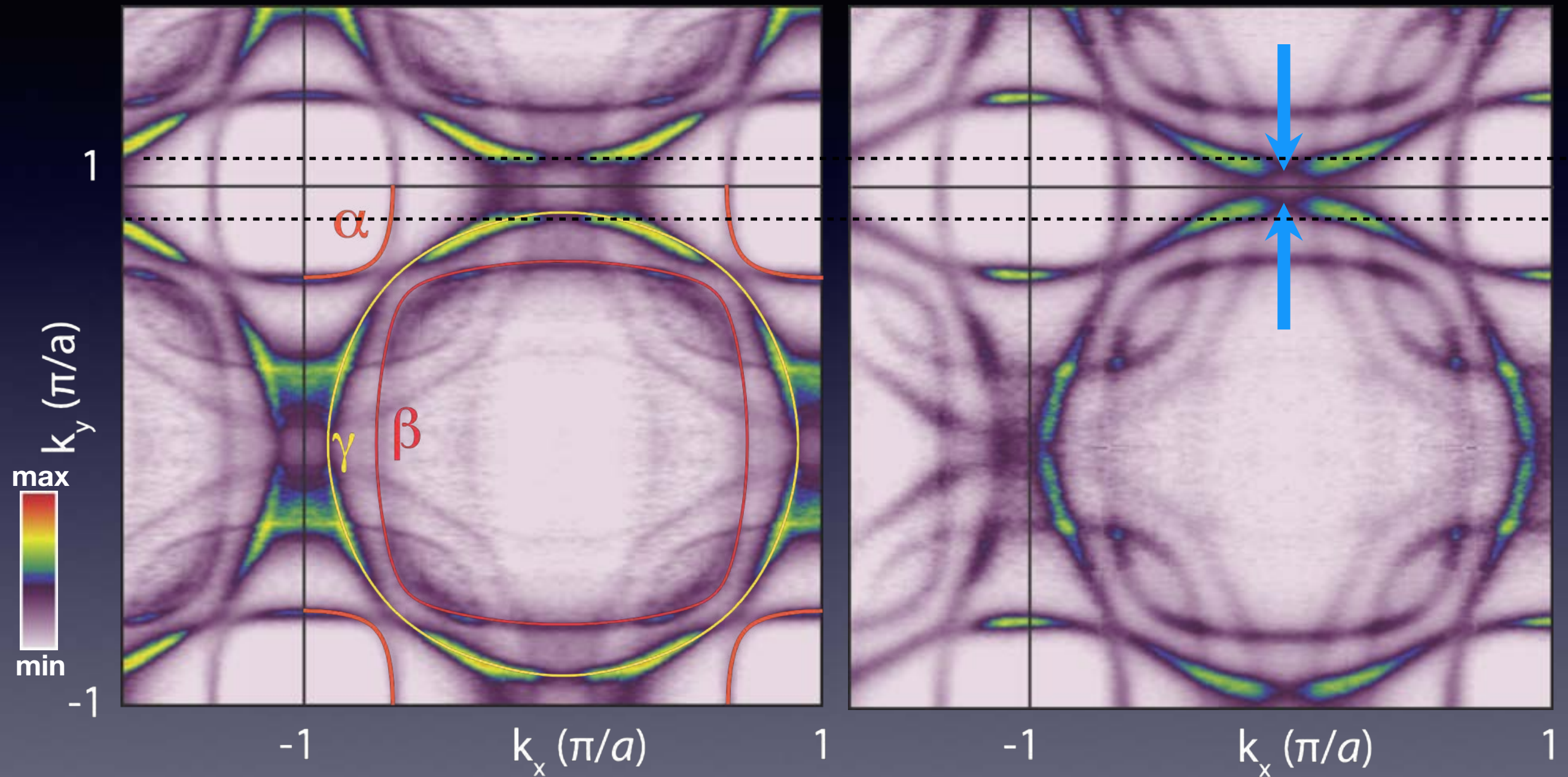
B. Burganov, et al., *Phys. Rev. Lett.* **116**, 197003

single crystal from A.P. Mackenzie

Can tensile strain push the van Hove singularity closer to E_F ?

Sr_2RuO_4 single crystal

Sr_2RuO_4 on SrTiO_3 (+0.9%)

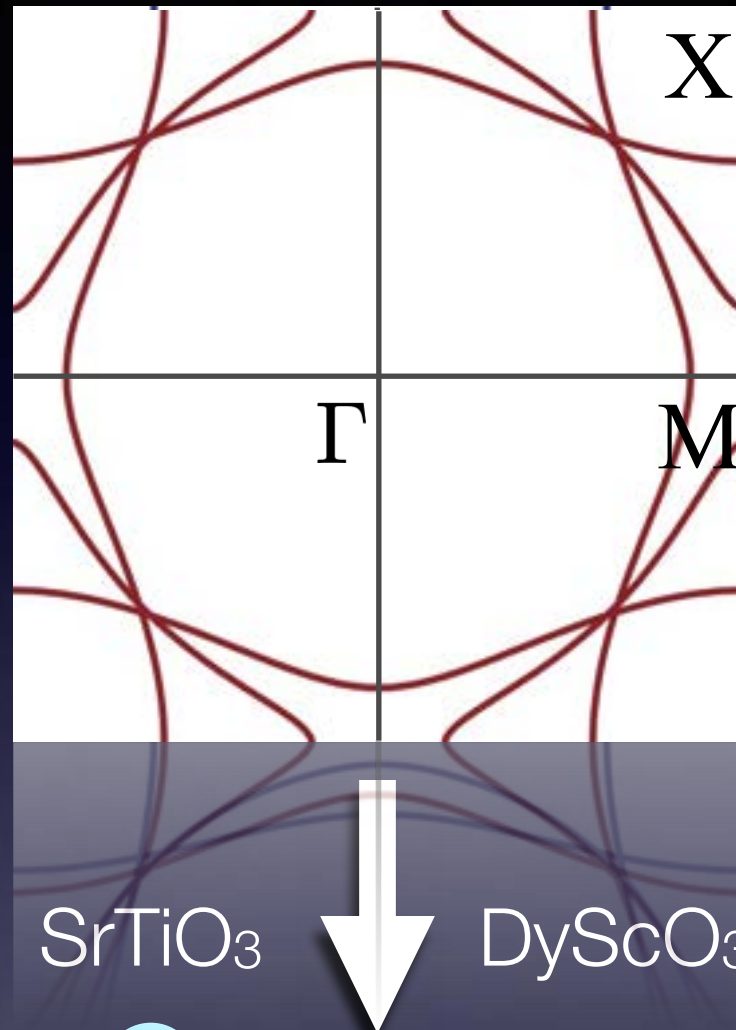


Epitaxial strain to enhance superconductivity in Sr_2RuO_4 ?

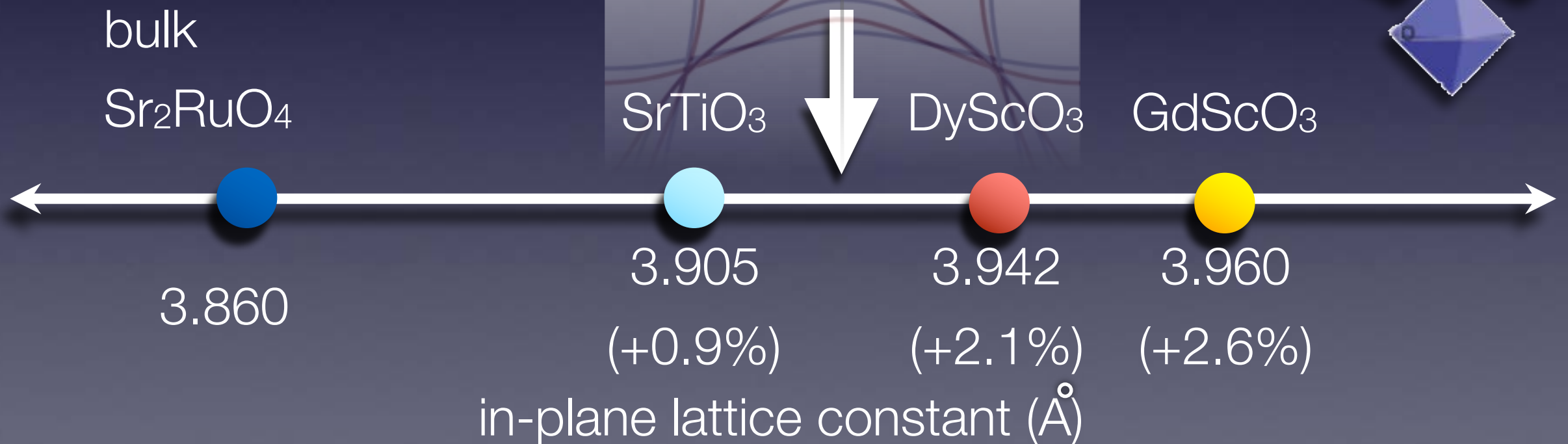
H							
Li	Be						
Na	Mg						
K	Ca	Sc	Ti	V	Cr	Mn	Fe
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru
Cs	Ba	La	Hf	Ta	W	Re	Os
Fr	Ra	Ac	Rf	Ha	106	107	108



Lifshitz transition



Ba_2RuO_4

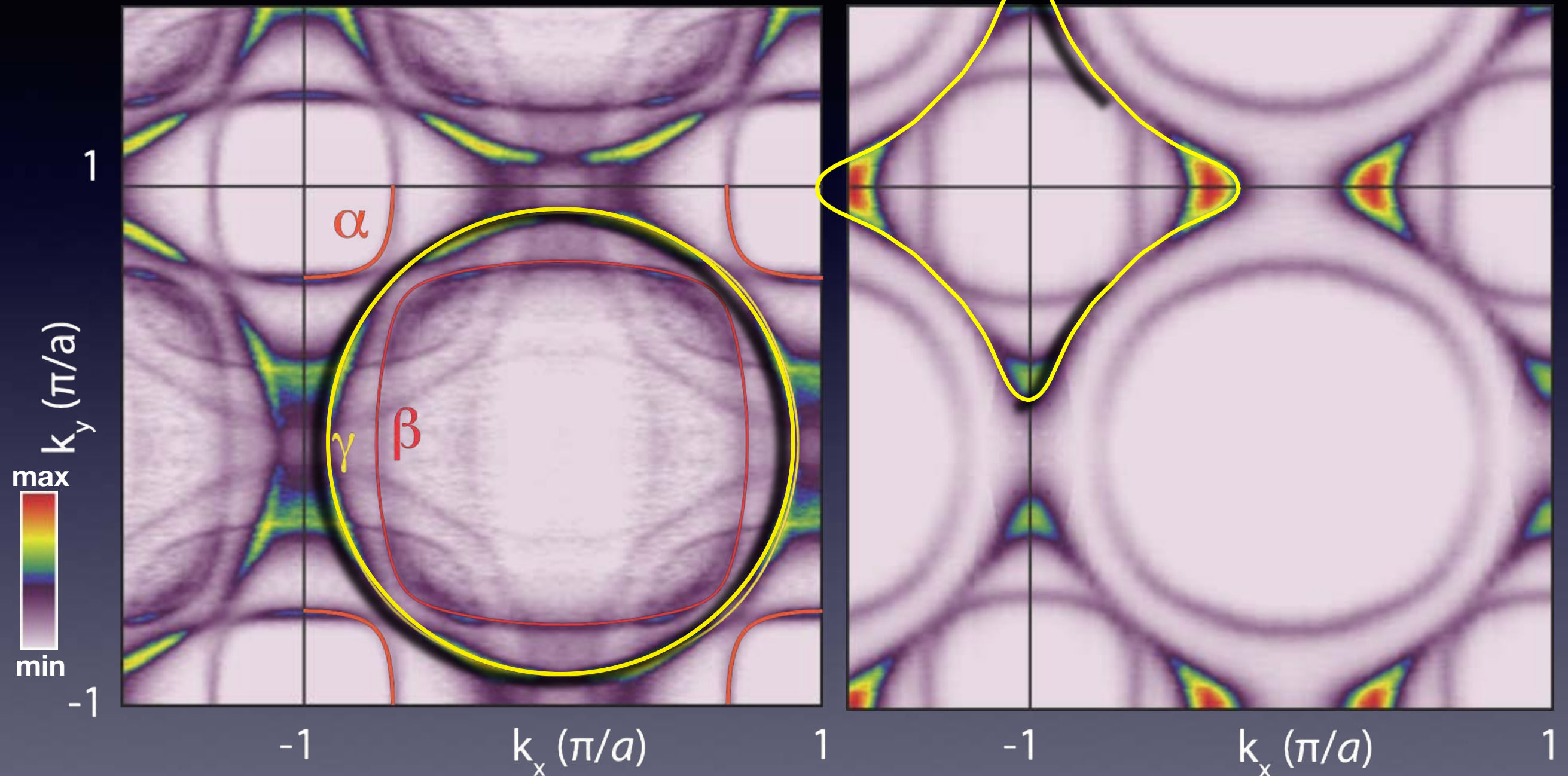


- Ba_2RuO_4 is metastable in bulk but can be epitaxially stabilized

Can tensile strain push the van Hove singularity closer to E_F ?

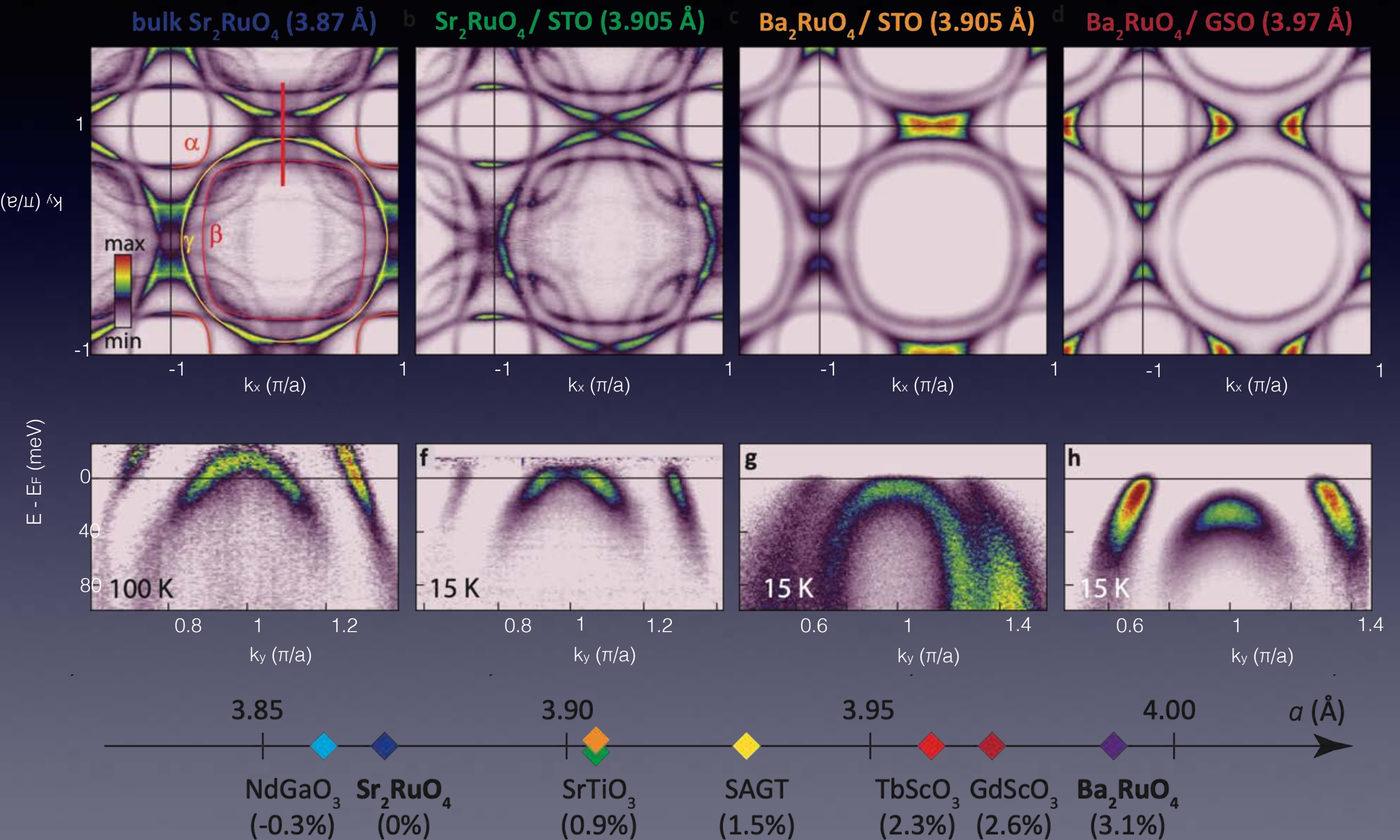
Sr_2RuO_4 single crystal

Ba_2RuO_4 on SrTiO_3 (+0.2%)

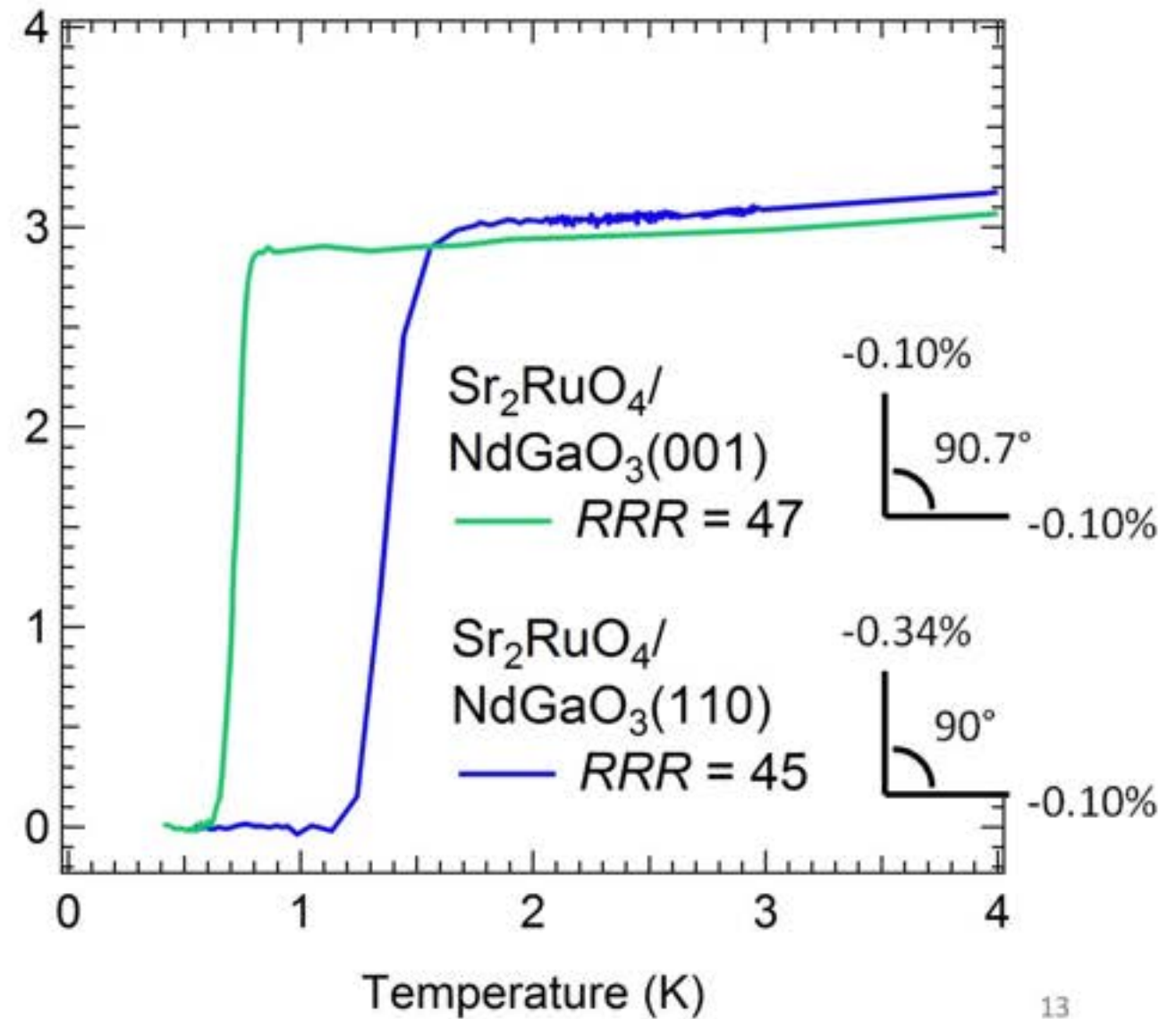
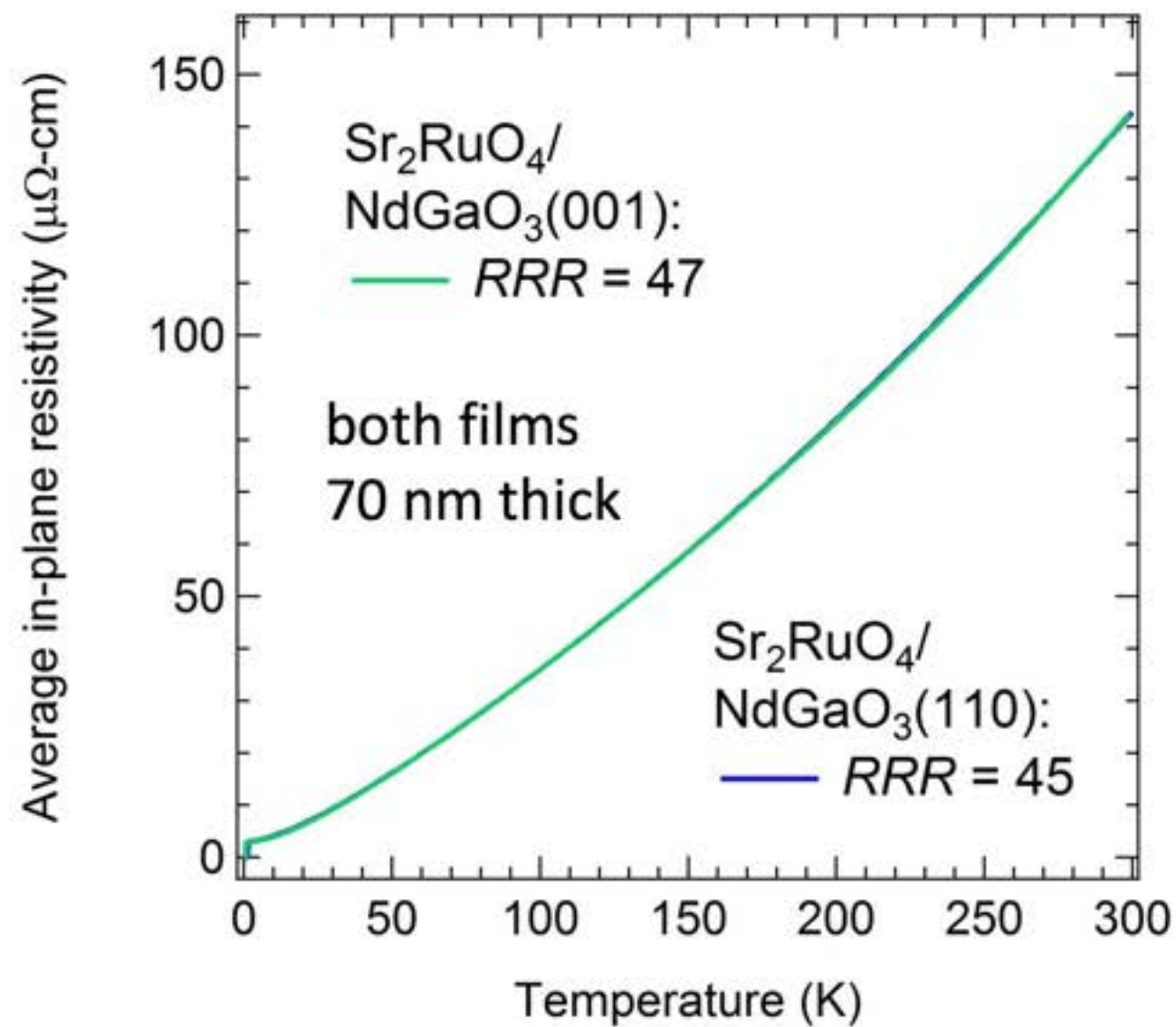


low T Hall coefficient changes sign from negative (Sr_2RuO_4) to positive (Ba_2RuO_4), consistent with ARPES

summary of Fermi surface & van Hove singularity evolution with strain



superconductivity depends on orientation of NdGaO₃ substrate



13

NdGaO₃ (110) Pbnm

NdGaO₃ (001) Pbnm

Darrell
Schlom

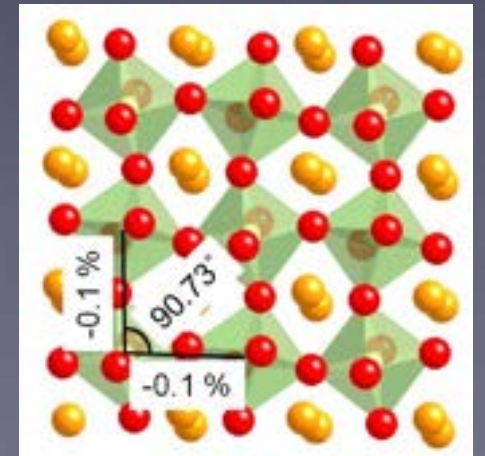
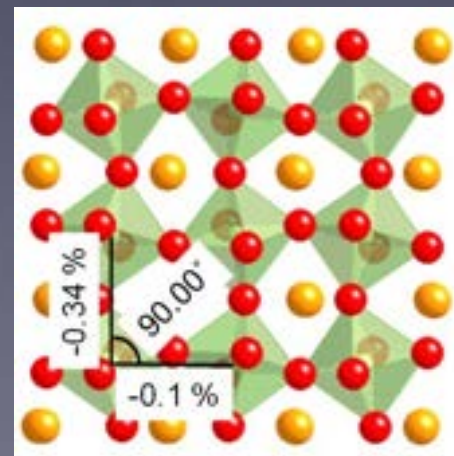
Hari
Nair

Jacob
Ruf

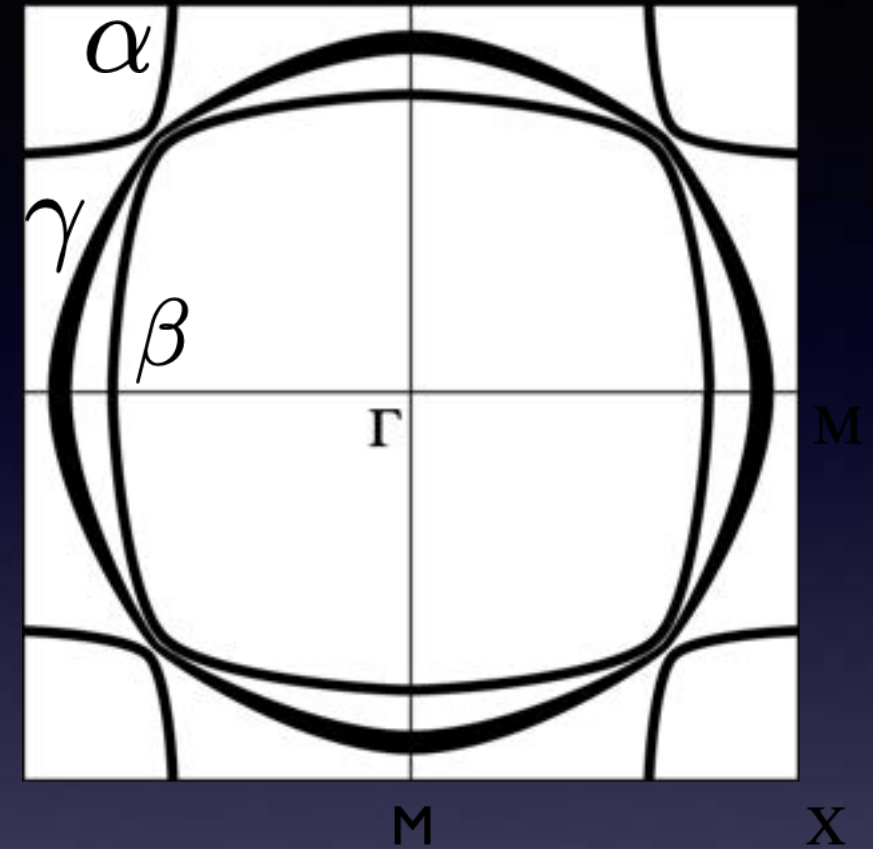
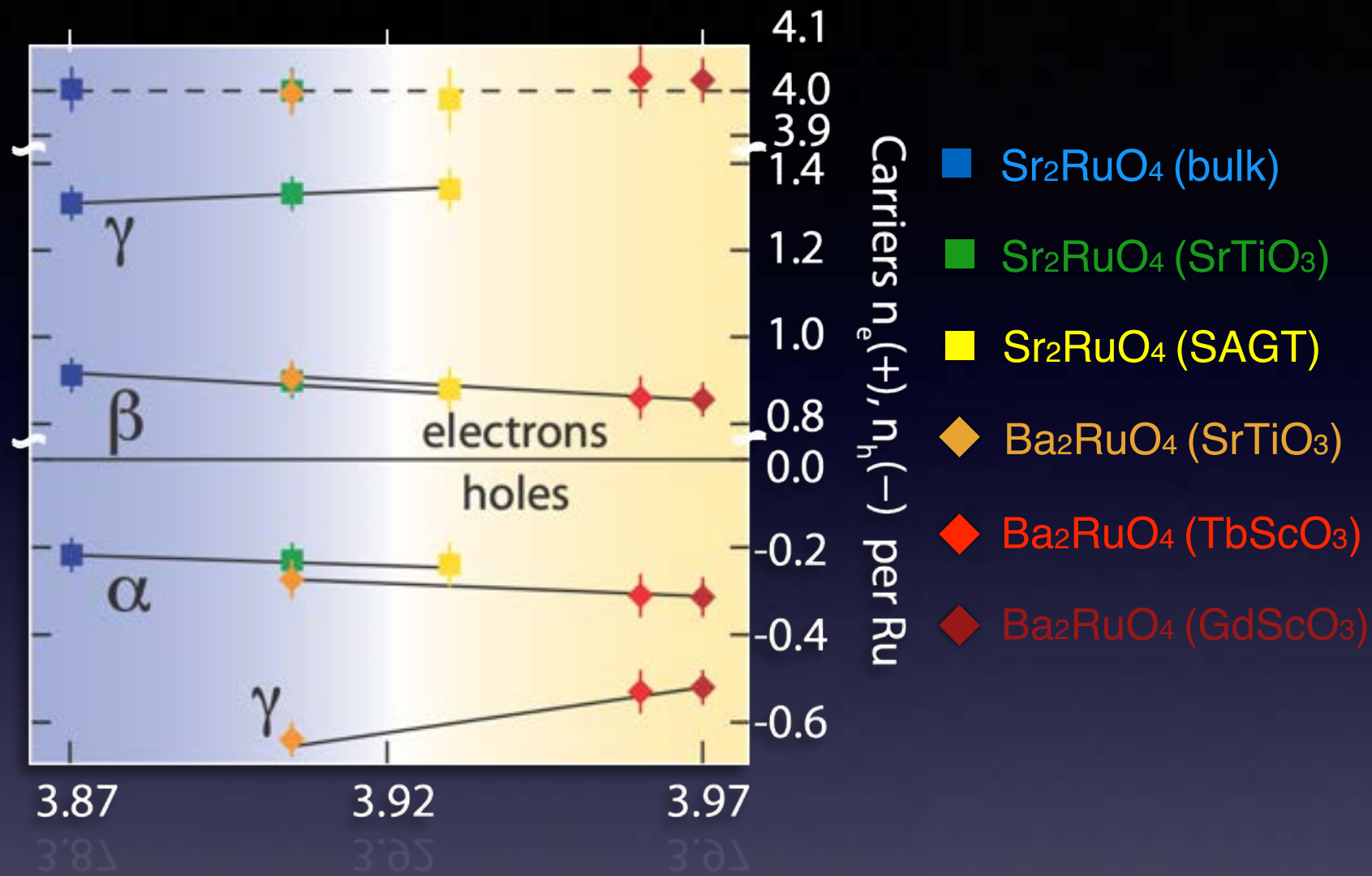
Nate
Schreiber



H.P. Nair, J.P. Ruf *et al.*, APL Matr. **6**, 101108



Detailed Luttinger count shows interorbital electron transfer



detailed Luttinger count shows that total number of electrons per Ru remains 4.00 ± 0.05 ; electrons are transferred from the 1D d_{yz} & d_{xz} bands to the d_{xy} band